

Non-perturbative foundation of generalized symmetries

—— *Wilsonian lattice regularized point of view* ——

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 Topology and symmetry in non-Abelian gauge theories are considered with Wilsonian lattice regularization. Recently, the concept of symmetry has been generalized; the important ingredients are given by higher-form, higher-group, and non-invertible symmetries. At first, we start from the continuum $SU(N)$ gauge theory with the 't Hooft twisted boundary condition, which is a traditional idea being equivalent to gauging the $\mathbb{Z}_N$ 1-form center symmetry. We provide a simpler explanation of van Baal's proof on the fractionality of the topological charge; its description is applicable to lattice gauge theory by generalizing Lüscher's construction of topology on the lattice. Thus we can recover the $SU(N)/\mathbb{Z}_N$ principal bundle from lattice $SU(N)$ gauge fields being subject to the $\mathbb{Z}_N$ -relaxed cocycle condition. We explicitly demonstrate the fractional topological charge, and verify an equivalence with other constructions reported recently based on different ideas. Gauging the $\mathbb{Z}_N$ 1-form center symmetry enables lattice gauge theories to couple with the $\mathbb{Z}_N$ 2-form gauge field as a simple lattice integer field, and to reproduce the Kapustin–Seiberg prescription in the continuum limit. Our construction is also applied to analyzing the higher-group and non-invertible symmetries in the $SU(N)$ gauge theory. Finally, we construct an axial $U(1)$ non-invertible symmetry on lattice chiral gauge theory with an anomalous matter content.

Acknowledgements

The present work is regarded as a part of our enterprise, fully regularized description of generalized symmetries with lattice regularization [1–8]. Originally, my aim has been to understand almost recent developments on generalized symmetries from a perspective of lattice gauge theory. To make my best endeavors to answer my questions (see main text), I have carried out these collaborated works with Motokazu Abe, Yamato Honda, Soma Onoda, Hiroshi Suzuki and Yuya Tanizaki; also Naoto Kan, Yuta Nagoya, and Hiroki Wada. I would like to especially thank them for gathering for my purpose.

Almost all parts of this paper is based on the following collaborated works [4, 8]:

- (1) M. Abe, O. Morikawa and S. Onoda, “Note on lattice description of generalized symmetries in $SU(N)/\mathbb{Z}_N$ gauge theories,” PRD **108**, 014506 (2023) [arXiv:2304.11813 [hep-th]].
- (2) Y. Honda, O. Morikawa, S. Onoda and H. Suzuki, “Lattice realization of the axial $U(1)$ non-invertible symmetry,” PTEP **2024**, no.4, 043B04 (2024) [arXiv:2401.01331 [hep-lat]].

In the first paper, I am the corresponding author. I contributed everything at all: motivation, calculation, and writing. Two students (Abe and Onoda) had checked the validity of my results.

The second paper is related to lattice realization of non-invertible symmetry. To consider it on the lattice, I was interested in the Harada–Tsutsui formalism [9] for anomalous gauge theories; in a quite similar way, Karasik [10] recently took non-invertible symmetry into account under the axial $U(1)$ transformation. Then I had studied some formulations of lattice anomalous gauge theories (e.g., Refs. [11, 12]), while Prof. Suzuki tried to construct a lattice chiral gauge formulation based on Lüscher [13]. We concluded that, to ensure the locality of lattice fermion integration measure, fermion fields couples to an external (non-dynamical) $U(1)$ gauge field in a chirally-asymmetric way, where the $U(1)$ sector possesses the gauge anomaly (see later). On the basis of such a construction, Honda, Onoda, Prof. Suzuki, and I realized the Karasik-type axial $U(1)$ non-invertible symmetry on the lattice. We also gave intrinsically 3-dimensional construction of the symmetry operator for rational rotation angles by employing a TQFT, the level- N BF theory (we are deeply indebted to Prof. Tanizaki for pointing this out to us).

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1 Introduction and summary

1.1 Symmetry and recent developments

The notion of gauge symmetry is one of the most impressive and fundamental structures in the established physics such as the Standard Model, general relativity, and also many phenomena in condensed matter physics. Enormous studies have focused on non-Abelian gauge fields with the gauge group $SU(N)$, because of its significant physical reality in quantum chromodynamics and also profound behaviors of strong-coupled and non-perturbative dynamics. Beyond perturbation theory at finite order in practice, one can observe nontrivial topological sectors, that is, instantons which are classified by the homotopy class $\pi_3(SU(N)) = \mathbb{Z}$ in four dimensions. Path integral formulation indicates that the summation over all topological sectors gives rise to a local and unitary quantum field theory.

A traditional gauge principle arises from the localization process of a global symmetry for phases (or elements of group) of matters [14]; this procedure is called gauging. A global symmetry in quantum field theory may be gauged as a probe, and then there is an 't Hooft anomaly when such a gauge symmetry is anomalous [15]. The 't Hooft anomalies in low and high energies should be identical because of its invariance under the renormalization group flow, which restricts the phase structure of strongly coupled gauge theories. The use of the so-called 't Hooft anomaly matching condition provides a notable approach to study the topological structure due to non-perturbative dynamics.

In the last decade, the concept of symmetry has been generalized [16, 17]. This so-called generalized global symmetry has been vigorously studied in not only particle physics but also condensed matter physics [18–23]. The important ingredients are as follows:

- higher-form symmetry: when a theory has symmetries acting on not only local operators but also a p -dimensional *charged* object, such a symmetry is called the p -form symmetry;
- higher-group symmetry: a categorical structure between some higher-form symmetries is realized [24, 25], where each symmetry cannot be gauged individually;
- non-invertible symmetry: a symmetry, which cannot be represented by a symmetry *group*, is given by a fusion rule between topological defects [10, 26–47]¹ so that there exist no inverse topological operators.

Examples of generalized symmetry would be given in string theory. For instance, the Green–Schwarz mechanism [53] is nothing but a higher-group such that for continuous 0-form and

¹ For recent works related to our approach in this paper, see also Refs. [39, 48–52].

1-form symmetries, the 2-form gauge field associated to the 1-form symmetry transforms under not only the 1-form gauge transformation but also the 0-form gauge transformation. The recent developments in line with these symmetries provide a quite different paradigm.

1.2 Brief review of coupling to higher-form gauge fields

It is well known that the $SU(N)$ gauge theory has the $\mathbb{Z}_N$ 1-form center symmetry. In order for non-Abelian gauge theories to couple with $\mathbb{Z}_N$ 2-form gauge fields associated with such 1-form symmetries, we can use the following procedure by Kapustin and Seiberg [16, 17, 54]: Let a topological field theory be described by a $\mathbb{Z}_N$ p -form gauge field. To represent this explicitly, introducing a $(p-1)$ -form compact scalar $B^{(p-1)}$ which satisfies $B^{(p-1)} \sim B^{(p-1)} + 2\pi$, a $U(1)$ p -form field $B^{(p)}$, and a Lagrange multiplier χ , we write the Lagrangian²

$$\chi \wedge \left(NB^{(p)} - dB^{(p-1)} \right). \quad (1.1)$$

The corresponding charged object behaves as

$$e^{i \int_{p\text{-cycle}} B^{(p)}} = e^{\frac{i}{N} \int_{p\text{-cycle}} dB^{(p-1)}} \in \mathbb{Z}_N. \quad (1.2)$$

Thus, the $\mathbb{Z}_N$ gauge field is written by the pair $(B^{(p)}, B^{(p-1)})$. Note that the $SU(N)$ gauge field A is blind to such a $\mathbb{Z}_N$ 2-form gauge field $(B^{(2)}, B^{(1)})$ because A (also the field strength F) is traceless. Now, *let us promote A to a $U(N)$ gauge field $\mathcal{A}$* . In the same way above, we constrain its $U(1)$ part as

$$\chi' \wedge \left(\text{tr } \mathcal{F} - dB^{(1)} \right). \quad (1.3)$$

This indicates that $U(1)$ is broken to $\mathbb{Z}_N$, that is, $U(N) = \frac{SU(N) \times U(1)}{\mathbb{Z}_N} \rightarrow SU(N)$. Further, by replacing $F \rightarrow \mathcal{F} - B^{(2)}$, the $SU(N)$ gauge field can couple to the $\mathbb{Z}_N$ 2-form gauge field; we have the $SU(N)/\mathbb{Z}_N$ gauge theory.

This recent prescription based on higher-form symmetries could be described equivalently by the $SU(N)/\mathbb{Z}_N$ gauge theory where the $SU(N)$ gauge fields obey the twisted boundary condition with the 't Hooft flux [55]. It is then known that the topological charge becomes fractional; in terms of the 2-form gauge field $B^{(2)}$, we see easily

$$\int d^4x \text{tr} \left(\mathcal{F} - B^{(2)} \right) \wedge \left(\mathcal{F} - B^{(2)} \right) = \int d^4x \text{tr } \mathcal{F} \wedge \mathcal{F} - N \int d^4x B^{(2)} \wedge B^{(2)}, \quad (1.4)$$

where the first term gives rise to an integer in the topological charge, and the second term can provide a fractional part because $\int_{2\text{-cycle}} B^{(2)} \in \frac{2\pi}{N} \mathbb{Z}$. The proof of the fractionality under

² We can regard χ as the vacuum expectation value (VEV) of a charge- N Higgs field H , and the compact scalar as the phase of H . If we take the $\chi \rightarrow \infty$ limit (or infinite Higgs VEV), we have the same structure.

the twisted boundary condition was done by van Baal [56], where the author constructs the transition function with the $\mathbb{Z}_N$ -center-valued cocycle condition and then identifies the topological classification of the $SU(N)$ principal bundle structure.

1.3 Topology in lattice gauge theory

Lüscher proved [57] that the $SU(N)$ principal bundle can be constructed from the $SU(N)$ lattice gauge theory (see also Refs. [58, 59]), while the topological structure on the lattice is nontrivial because the discretization of the spacetime breaks its continuity. We can recover this “continuity” by Lüscher’s construction, and then, classify the integer topological charges. Recall that quantum field theory, a physical system with an infinite number of degrees of freedom, would be mathematically not well-defined as it stands, and the lattice regularization is the most well-developed non-perturbative framework. This work is quite awesome since it provides a solid foundation on our understanding of topological classifications, which enrich the nontrivial dynamics of gauge fields (e.g., the index theorem [60, 61] and so on).

Recently, in Refs. [1, 3], by generalizing this, I and the collaborators had constructed the non-simply-connected $U(1)/\mathbb{Z}_N$ or $SU(N)/\mathbb{Z}_N$ principal bundle in the lattice theory coupled with the $\mathbb{Z}_N$ 2-form gauge field, and proved the fractionality of the topological charge on the lattice. These studies have achieved the fully regularized framework on the modern viewpoint of quantum field theories with generalized symmetries.

The proof in Ref. [3] is given in a sophisticated way based on the principle of the locality, $SU(N)$ gauge invariance, and $\mathbb{Z}_N$ 1-form gauge invariance, while this idea looks different from that for the $U(1)$ case in Ref. [1] and the construction is quite complicated. Thus, from the analytical viewpoint in lattice gauge theory, it may be hard to explicitly demonstrate the traditional knowledge on the ’t Hooft twisted boundary condition, and recent developments of non-Abelian gauge theories with higher-form symmetries. Also the relation between the Kapustin–Seiberg prescription by the $U(1)$ fields and the lattice construction by the 2-form integer lattice field is not obvious; it is puzzling how to take the continuum limit of an integer field on the lattice so that its field configuration is smooth.

1.4 Constructive proof and application to other generalized symmetries

In this paper, we reconstruct the $SU(N)/\mathbb{Z}_N$ principal bundle from the lattice $SU(N)$ gauge theory with the twisted boundary condition. First, in Sect. 2, we start from the continuum theory following van Baal [56], on which the construction of the $U(1)/\mathbb{Z}_N$ principal bundle [1] is based. We can make his discussions much simpler owing to the extension of $SU(N)$ to $U(N)$ like as the Kapustin–Seiberg prescription. Actually, the original proof seems to be incompatible with interpolated transition functions written by lattice $SU(N)$ gauge

fields, but the above description is applicable to both continuum and lattice. In the next section, we give the lattice realization of it, by using the generalized Lüscher’s construction [3]. We can then show how to establish an equivalence between the constructions in this paper and Refs. [1, 3]. The distinguishing feature is that our construction provides concrete expressions for $\mathcal{F}$ and $B^{(2)}$ defined on the lattice while those are not defined in Ref. [3]. That is, this point is not necessary to prove the fractionality of the topological charge, but we can see the fractional structure as an explicit form and apply this construction to some related issues.

As an important perspective from our construction, in Sect. 4, we perform the $U(1)$ 1-form gauge transformation, to which the gauged $\mathbb{Z}_N$ 1-form center symmetry is promoted. We then directly reproduce the Kapustin–Seiberg prescription in terms of lattice fields. Section 5 shows applications of our expressions to analyzing higher-group and non-invertible symmetries: We provide a lattice description of the higher-group structure in the $SU(N)$ gauge theory with the so-called instanton-sum modification [62], which is a restriction of the topological sectors to the instanton numbers characterized by an integer without violating the locality [63–66]. We briefly discuss a non-invertible condensation defect among $SU(N)$.

The non-invertible symmetry is still developing and deserve further investigations, and so we apply our approach to such recent developments. Note that it is also interesting to consider the case with matter fields, especially lattice fermions, whose construction is nontrivial in lattice gauge theory. Quite recently, it was found [40, 41] that, in a system with the ABJ anomaly [67, 68], the axial $U(1)$ transformation possesses the non-invertible symmetry if rotation angles are rational. The crucial observation is given as a gauge-invariant form of the fractional-level Chern–Simons action on a 3-dimensional closed surface, which is coupled with a topological field theory, the fractional quantum Hall state [69]. To implement it on the lattice is a very hard problem because of the following issues: the lattice Chern–Simons term³, lattice chiral gauge theory, and anomalous gauge theory. In Sect. 6, employing the lattice formulation of the $U(1)$ chiral gauge theory [13] based on the Ginsparg–Wilson relation [71–78], and using the axial $U(1)$ anomaly with finite lattice spacings [79, 80], we make an attempt to realize a lattice axial $U(1)$ non-invertible symmetry. We focus on the prescription of Ref. [10]⁴ for arbitrary rotation angles, and a lattice $\mathbb{Z}_N$ topological field

³ In the case of the $U(1)$ gauge theory, one may use the modified Villain formulation of the Chern–Simons theory [70], where the lattice degrees of freedom are described in terms of non-compact $U(1)$ variables, that is, the *gauge potential* directly. An extendable approach for non-Abelian gauge theories, however, can be provided by the Wilson-type formulation that use compact gauge variables as fundamental variables.

⁴ See also Ref. [43] for an analogous construction.

theory for fractional rotation angles in the unit of $1/N$. In the latter one, we evaluate fusion rules of symmetry operators with lattice regularization.

We expect that the procedure in this paper is applicable broadly as an underlying fully regularized framework.

2 A revision of the $SU(N)$ gauge theory with twisted boundary condition

At first, we review how the topological charge becomes fractional in the $SU(N)$ gauge theory with a twisted boundary condition on a torus. The original proof by van Baal [56] is modified in a simpler way, in accordance with the recent viewpoint on non-Abelian gauge theories coupled with $\mathbb{Z}_N$ gauge fields [54].

A four-dimensional periodic torus with the size L is give by $T^4 \equiv \{x \in \mathbb{R}^4 \mid 0 \leq x_\mu < L \text{ for all } \mu\}$ with the identification $x_\mu + L \sim x_\mu$, where μ runs over 1, 2, 3, and 4. We impose the twisted boundary condition for the gauge field $A_\mu(x) \in \mathfrak{su}(N)$,

$$A_\mu(x_\nu + L, x_{\lambda \neq \nu}) = h_\nu(x)^{-1} A_\mu(x) h_\nu(x) - i h_\nu(x)^{-1} \partial_\mu h_\nu(x), \quad (2.1)$$

with the $\mathbb{Z}_N$ -relaxed cocycle condition

$$\begin{aligned} & h_\mu(x_\mu = 0, x_\nu = L, x_{\lambda \neq \mu, \nu})^{-1} h_\nu(x_\mu = x_\nu = 0, x_{\lambda \neq \mu, \nu})^{-1} \\ & \times h_\mu(x_\mu = x_\nu = 0, x_{\lambda \neq \mu, \nu}) h_\nu(x_\mu = L, x_\nu = 0, x_{\lambda \neq \mu, \nu}) = e^{\frac{2\pi i}{N} z_{\mu\nu}} \in \mathbb{Z}_N. \end{aligned} \quad (2.2)$$

Here $z_{\mu\nu} \in \mathbb{Z}$, which is anti-symmetric as $z_{\mu\nu} = -z_{\nu\mu}$, stands for the 't Hooft flux [55].

The topological charge is specified by the transition function $h_\mu(x)$. We see the Lemma proved by van Baal and Lüscher [56, 57] as follows:

Lemma. *Subject to the twisted boundary condition (2.2) (or the periodic one with $z_{\mu\nu} = 0$), the topological charge is written, in terms of $h_\mu(x)$, by*

$$\begin{aligned} Q &= \frac{1}{32\pi^2} \int_{T^4} d^4x \sum_{\mu, \nu, \rho, \sigma} \epsilon_{\mu\nu\rho\sigma} \text{tr} [F_{\mu\nu}(x) F_{\rho\sigma}(x)] \\ &= -\frac{1}{24\pi^2} \sum_{\mu, \nu, \rho, \sigma} \epsilon_{\mu\nu\rho\sigma} \left\{ 3 \int dx_\rho dx_\sigma \text{tr} \left[(h_\mu \partial_\rho h_\mu^{-1})_{x_\mu=x_\nu=0} (h_\nu^{-1} \partial_\sigma h_\nu)_{x_\mu=L, x_\nu=0} \right] \right. \\ &\quad \left. + \int dx_\nu dx_\rho dx_\sigma \text{tr} \left[(h_\mu^{-1} \partial_\nu h_\mu) (h_\mu^{-1} \partial_\rho h_\mu) (h_\mu^{-1} \partial_\sigma h_\mu) \right]_{x_\mu=0} \right\}. \end{aligned} \quad (2.3)$$

If we have no twists ($z_{\mu\nu} = 0$), $Q \in \mathbb{Z}$ characterizes the homotopy type $\pi_3(SU(N)) = \mathbb{Z}$. On the other hand, turning on $z_{\mu\nu}$, we can see that Q becomes fractional by considering an

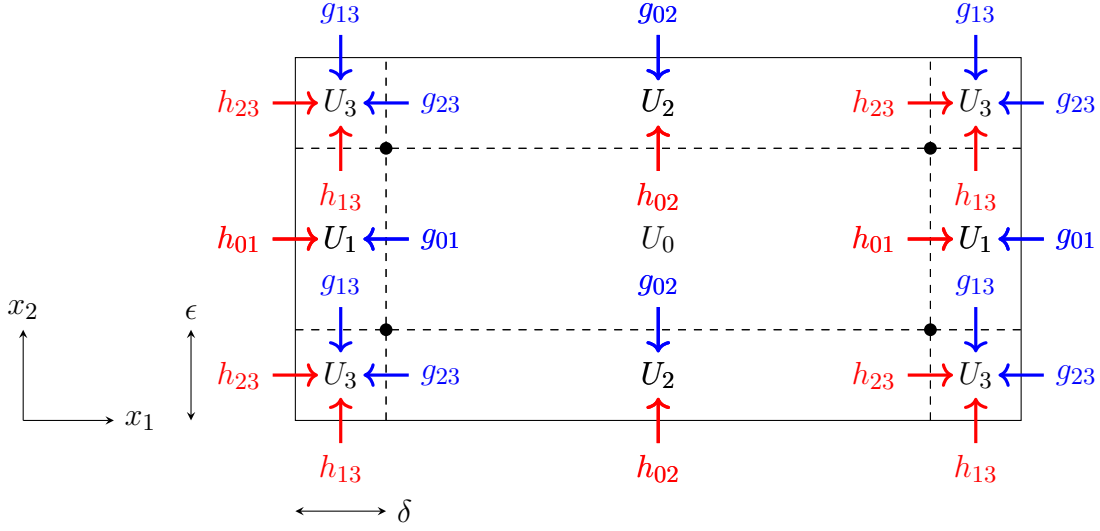


Fig. 1 The structure of the fiber bundle of the base space T^2 . $\{U_i\}$ is a set of patches of T^2 , where U_0 is the main region, U_μ is near the boundary at $x_\mu = L$, and U_3 is the corner.

appropriate fiber bundle structure due to the first homotopy group $\pi_1(SU(N)/\mathbb{Z}_N) = \mathbb{Z}_N$ on T^4 . To see this, for simplicity, we consider the bundle structure of T^2 as depicted in Fig. 1. Letting U_i with $\epsilon > 0$ and $\delta > 0$ be a covering of T^2 , h_{ij} and g_{ij} are transition functions at $x \in U_i \cap U_j$ such that $A_\mu(U_i) \rightarrow A_\mu(U_j)$ in the positive and negative x directions, respectively. In the limit of $\epsilon \rightarrow 0$ and $\delta \rightarrow 0$, the patches $U_{i \neq 0}$ shrink and the nontrivial transition indicates that $A_\mu(U_0) \rightarrow A_\mu(U_\nu) \rightarrow A_\mu(U_0)$, that is, $A_\mu(x_\nu + L, x_{\lambda \neq \nu}) \mapsto A_\mu(x)$. Therefore, on T^4 , we have $h_\mu = h_{0\mu} g_{0\mu}^{-1}$ with an appropriate assignment of indices.

Now, following the van Baal's prescription [56], we introduce the loop factor $\tilde{\zeta}_\mu(x)$ in terms of the Cartan subalgebra of $SU(N)$,

$$\tilde{\zeta}_\mu(x) \equiv \exp \left(-\frac{2\pi i}{N} \sum_{\nu > \mu} \frac{z_{\mu\nu} x_\nu}{L} T_1 \right), \quad (2.4)$$

where T_1 is a generator of $SU(N)$, $T_1 \equiv \text{diag}(1, 1, \dots, 1, -N + 1)$. Then we find

$$\begin{aligned} & \tilde{\zeta}_\mu(x_\mu = 0, x_\nu = L, x_{\lambda \neq \mu, \nu})^{-1} \tilde{\zeta}_\nu(x_\mu = x_\nu = 0, x_{\lambda \neq \mu, \nu})^{-1} \\ & \times \tilde{\zeta}_\mu(x_\mu = x_\nu = 0, x_{\lambda \neq \mu, \nu}) \tilde{\zeta}_\nu(x_\mu = L, x_\nu = 0, x_{\lambda \neq \mu, \nu}) = e^{\frac{2\pi i}{N} z_{\mu\nu}} \in \mathbb{Z}_N. \end{aligned} \quad (2.5)$$

This cocycle condition is identical to that of $h_\mu(x)$ (2.2), and thus, $\tilde{\zeta}_\mu(x)$ possesses the same nontrivial “winding” modulo 1 as the original system.

Theorem. Given $\forall h_\mu(x)$, we can write

$$h_\mu(x) = h_{0\mu} g_{0\mu}^{-1} = \hat{h}_{0\mu} \tilde{\varsigma}_\mu \hat{g}_{0\mu}^{-1}, \quad (2.6)$$

where we have introduced

$$\tilde{\varsigma}_\mu = \tilde{h}_{0\mu} \tilde{g}_{0\mu}^{-1}, \quad \hat{h}_{0\mu} = h_{0\mu} \tilde{h}_{0\mu}^{-1}, \quad \hat{g}_{0\mu} = g_{0\mu} \tilde{g}_{0\mu}^{-1}, \quad (2.7)$$

and $\hat{h}_{0\mu}$, $\hat{g}_{0\mu}$ are defined in terms of the transition function $\hat{h}_\mu(x) = \hat{h}_{0\mu} \hat{g}_{0\mu}^{-1}$ obeying the periodic boundary condition.

Corollary. Let $Q[\cdot]$ be a mapping from a transition function to the topological charge given by the above Lemma. Then, we have $Q[h] = Q[\tilde{\varsigma}] + Q[\hat{h}]$.

The proof being given in Ref. [56], we can then show that $Q[\tilde{\varsigma}] \in \frac{1}{N}\mathbb{Z}$ and $Q[\hat{h}] \in \mathbb{Z}$. Thus the total topological charge on the $SU(N)/\mathbb{Z}_N$ principal bundle can be fractional.

The above original construction is quite complicated because its description enjoys the $SU(N)$ structure at any stage of computations. Also, lattice gauge theory as we will describe later seems to be incompatible with the above Theorem. If such a difficulty of $SU(N)$ keeps us in mind of the recent development about the generalized symmetry [54], one may begin with the loop factor $\varsigma_\mu(x) \in U(1)$ redefined by

$$\varsigma_\mu(x) \equiv \exp \left(-\frac{2\pi i}{N} \sum_{\nu > \mu} \frac{z_{\mu\nu} x_\nu}{L} \right), \quad (2.8)$$

and the $U(N)$ structure which is combined with $\varsigma_\mu(x)$ into $SU(N)$ -valued results. This definition of $\varsigma_\mu(x)$ again satisfies the cocycle condition (2.5). By using $\varsigma_\mu(x) \in U(1)$, let us define

$$\check{h}_\mu(x) \equiv \varsigma_\mu(x)^{-1} h_\mu(x) \in U(N). \quad (2.9)$$

Then, we immediately find that

$$\begin{aligned} & \check{h}_\mu(x_\mu = 0, x_\nu = L, x_{\lambda \neq \mu, \nu})^{-1} \check{h}_\nu(x_\mu = x_\nu = 0, x_{\lambda \neq \mu, \nu})^{-1} \\ & \times \check{h}_\mu(x_\mu = x_\nu = 0, x_{\lambda \neq \mu, \nu}) \check{h}_\nu(x_\mu = L, x_\nu = 0, x_{\lambda \neq \mu, \nu}) = 1. \end{aligned} \quad (2.10)$$

$\check{h}_\mu(x)$ denotes the transition function of the “ $U(N)$ principal bundle” with an integer 2nd Chern number.

Substituting $h_\mu(x) = \varsigma_\mu(x) \check{h}_\mu(x)$ to the expression of the topological charge, and using the integer topological charge $\check{Q}$ by $\check{h}_\mu(x)$ instead of $h_\mu(x)$ in the above expression of Q ,⁵ we

⁵ This integer topological charge for the $U(N)$ gauge field, $\int \text{tr } F \wedge F$, is somewhat different from the definition of the 2nd Chern class, $\int (\text{tr } F \wedge F - \text{tr } F \wedge \text{tr } F)$, up to integers.

have

$$\begin{aligned}
Q &= \frac{1}{8N} \sum_{\mu, \nu, \rho, \sigma} \epsilon_{\mu\nu\rho\sigma} z_{\mu\nu} z_{\rho\sigma} + \check{Q} \\
&\quad - \frac{i}{4\pi N} \sum_{\mu, \nu, \rho > \mu, \sigma} \epsilon_{\mu\nu\rho\sigma} \frac{z_{\mu\rho}}{L} \int dx_\rho dx_\sigma \operatorname{tr} \left[(\check{h}_\nu^{-1} \partial_\sigma \check{h}_\nu)_{x_\mu=L, x_\nu=0} - (\check{h}_\nu \partial_\sigma \check{h}_\nu^{-1})_{x_\mu=x_\nu=0} \right] \\
&= -\frac{1}{8N} \sum_{\mu, \nu, \rho, \sigma} \epsilon_{\mu\nu\rho\sigma} z_{\mu\nu} z_{\rho\sigma} + \check{Q}.
\end{aligned} \tag{2.11}$$

In the last line, we have used, for $\mu < \nu$,

$$0 = \operatorname{tr} (h_\mu^{-1} \partial_\nu h_\mu) = -2\pi i \frac{z_{\mu\nu}}{L} + \operatorname{tr} (\check{h}_\mu^{-1} \partial_\nu \check{h}_\mu). \tag{2.12}$$

The first term in Q is fractional, that is, it is proved that $Q \in \frac{1}{N}\mathbb{Z}$; the fractional part and $\check{Q} \in \mathbb{Z}$ are naturally decomposed.

3 Lattice $SU(N)$ gauge theory with twisted boundary condition

3.1 Setup as a lattice formulation

It would be difficult to consider the topological structure in lattice gauge theories, where the spacetime is discretized into a set of lattice points and then its continuity is broken. We can recover this feature by the Lüscher's construction of the $SU(N)$ principal bundle from the lattice $SU(N)$ gauge theory [57]. Quite recently, this construction has been generalized to non-Abelian gauge theories coupled with $\mathbb{Z}_N$ gauge fields [3]. Let us demonstrate the $SU(N)/\mathbb{Z}_N$ principal bundle in line with the van Baal's proof in the previous section.

The lattice $\Gamma \equiv \{n \in \mathbb{Z}^4 \mid 0 \leq n_\mu < L \text{ for all } \mu\}$ divides T^4 into hypercubes $c(n)$ as

$$c(n) \equiv \{x \in \mathbb{R}^4 \mid 0 \leq (x_\mu - n_\mu) \leq 1 \text{ for all } \mu\}. \tag{3.1}$$

We also define the boundary of two hypercubes, called the face, $f(n, \mu) \equiv c(n - \hat{\mu}) \cap c(n)$, where $\hat{\mu}$ is a unit vector in the positive μ direction, and a two-dimensional plaquette as the intersection of four hypercubes: $p(n, \mu, \nu) \equiv c(n) \cap c(n - \hat{\mu}) \cap c(n - \hat{\nu}) \cap c(n - \hat{\mu} - \hat{\nu})$. Our lattice setup is summarized in Fig. 2.

Suppose that the link variable $U(n, \mu) \in SU(N)$, which lives on the link connecting n and $n + \hat{\mu}$, obeys the twisted boundary condition,

$$U(n + L\hat{\nu}, \mu) = g_\nu(n)^{-1} U(n, \mu) g_\nu(n + \hat{\mu}). \tag{3.2}$$

The cocycle condition is given by

$$g_\mu(n + L\hat{\nu})^{-1} g_\nu(n)^{-1} g_\mu(n) g_\nu(n + L\hat{\mu}) = e^{\frac{2\pi i}{N} z_{\mu\nu}} \in \mathbb{Z}_N. \tag{3.3}$$

This represents the 't Hooft flux on the lattice.

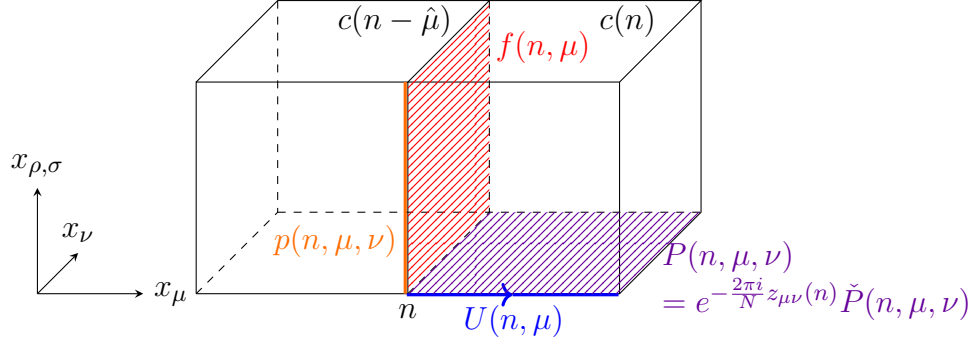


Fig. 2 Lattice setup and plaquettes for the twisted and periodic link variables. To illustrate generic variables, we use a lattice field $z_{\mu\nu}(n) = z_{\mu\nu}\delta_{n_\mu, L-1}\delta_{n_\nu, L-1}$ as we will define later.

To rewrite the link variable $U(n, \mu)$ in terms of the periodic one, we assume that $U(n, \mu)$ is defined on $0 \leq n_{\nu \neq \mu} \leq L$ and $0 \leq n_\mu \leq L-1$. We define the periodic variable $\check{U}(n, \mu)$ by

$$U(n, \mu) = \begin{cases} \check{U}(n, \mu)g_\mu(n) & \text{for } n_\mu = L-1, \\ \check{U}(n, \mu) & \text{otherwise.} \end{cases} \quad (3.4)$$

One can find that

$$\begin{aligned} & g_\mu(n_\nu = L-1)U(n_\mu = L, n_\nu = L-1, \nu)U(n_\mu = L-1, n_\nu = L, \mu)^{-1}g_\nu(n_\mu = L-1)^{-1} \\ & = e^{-\frac{2\pi i}{N}z_{\mu\nu}}\check{U}(n_\mu = 0, n_\nu = L-1, \nu)\check{U}(n_\mu = L-1, n_\nu = 0, \mu)^{-1}. \end{aligned} \quad (3.5)$$

This shows that, as depicted in Fig. 2, the plaquette $P(n, \mu, \nu)$,

$$P(n, \mu, \nu) \equiv U(n, \mu)U(n + \hat{\mu}, \nu)U(n + \hat{\nu}, \mu)^{-1}U(n, \nu)^{-1}, \quad (3.6)$$

can be written by $P(n, \mu, \nu) = e^{-\frac{2\pi i}{N}z_{\mu\nu}}\check{P}(n, \mu, \nu)$ up to gauge functions if this Wilson loop passes the corner of the lattice, where $\check{P}(n, \mu, \nu)$ is the plaquette constructed by $\check{U}(n, \mu)$; otherwise, $P(n, \mu, \nu) = \check{P}(n, \mu, \nu)$. In what follows, we simply say $P(n, \mu, \nu) = e^{-\frac{2\pi i}{N}z_{\mu\nu}}\check{P}(n, \mu, \nu)$ as this meaning.

3.2 $SU(N)/\mathbb{Z}_N$ principal bundle and fractionality on the lattice

Let us apply the Lüscher's construction of the transition function to $U(n, \mu)$ and $\check{U}(n, \mu)$, whose transition functions are denoted by $v_{n, \mu}(x)$ and $\check{v}_{n, \mu}(x)$, respectively. We aim to write $v_{n, \mu}(x)$ in terms of $\check{v}_{n, \mu}(x)$.

3.2.1 Lüscher's interpolation functions

A transition function $g_{n,\mu}(u; x)$ for a link variable $u(n, \mu)$ at $\forall x \in f(n, \mu)$ is defined in terms of the Lüscher's interpolation function $S_{n,\mu}^m(u; x)$ [57]. We summarize the Lüscher's construction for the transition function [3, 57]. First, we define new link variables which are constructed by

$$\begin{cases} u_{xy}^n = w^n(x)u(x, \mu)w^n(y) & \text{if } y = x + \hat{\mu}, \\ u_{xy}^n = (u_{yx}^n)^{-1} & \text{if } y = x - \hat{\mu}, \end{cases} \quad (3.7)$$

where the standard parallel transporter $w^n(u; x)$ at the corners of $f(n, \mu)$ is given by

$$w^n(u; x) = u(n, 4)^{z_4} u(n + z_4 \hat{4}, 3)^{z_3} u(n + z_4 \hat{4} + z_3 \hat{3}, 2)^{z_2} u(n + z_4 \hat{4} + z_3 \hat{3} + z_2 \hat{2}, 1)^{z_1} \quad \text{for } z_\mu \equiv x_\mu - n_\mu = 0 \text{ or } 1. \quad (3.8)$$

By these link variables, we can construct the transition function at the corners of $f(n, \mu)$,

$$u_{xy}^{n-\hat{\mu}} = v_{n,\mu}(x)u_{xy}^n v_{n,\mu}(y)^{-1}, \quad (3.9)$$

$$v_{n,\mu} \equiv w^{n-\hat{\mu}}(x)w^n(x)^{-1}. \quad (3.10)$$

Next, in order to calculate the topological charge which is similar to the above Lemma, we interpolate the transition function to $\forall x \in f(n, \mu)$. Then, we need to define the interpolation function $S_{n,\mu}^m$. We first label the corners $\{s_i\}_{i=0,\dots,7}$ of $f(n, \mu)$ as follows: letting $\alpha, \beta, \gamma \in \{1, 2, 3, 4\} \setminus \{\mu\}$ and $\alpha < \beta < \gamma$,

$$\begin{aligned} s_0 &= n, & s_1 &= n + \hat{\alpha}, & s_2 &= n + \hat{\beta}, & s_3 &= n + \hat{\gamma}, \\ s_4 &= n + \hat{\alpha} + \hat{\beta} + \hat{\gamma}, & s_5 &= n + \hat{\alpha} + \hat{\gamma}, & s_6 &= n + \hat{\alpha} + \hat{\beta}, & s_7 &= n + \hat{\beta} + \hat{\gamma}. \end{aligned} \quad (3.11)$$

Then, for $m = n, n - \hat{\mu}$, we define the interpolation functions,

$$f_{n,\mu}^m(x_\gamma) = (u_{s_3 s_0}^m)^{y_\gamma} (u_{s_0 s_3}^m u_{s_3 s_7}^m u_{s_7 s_2}^m u_{s_2 s_0}^m)^{y_\gamma} u_{s_0 s_2}^m (u_{s_2 s_7}^m)^{y_\gamma}, \quad (3.12)$$

$$g_{n,\mu}^m(x_\gamma) = (u_{s_5 s_1}^m)^{y_\gamma} (u_{s_1 s_5}^m u_{s_5 s_4}^m u_{s_4 s_6}^m u_{s_6 s_1}^m)^{y_\gamma} u_{s_1 s_6}^m (u_{s_6 s_4}^m)^{y_\gamma}, \quad (3.13)$$

$$h_{n,\mu}^m(x_\gamma) = (u_{s_3 s_0}^m)^{y_\gamma} (u_{s_0 s_3}^m u_{s_3 s_5}^m u_{s_5 s_1}^m u_{s_1 s_0}^m)^{y_\gamma} u_{s_0 s_1}^m (u_{s_1 s_5}^m)^{y_\gamma}, \quad (3.14)$$

$$k_{n,\mu}^m(x_\gamma) = (u_{s_7 s_2}^m)^{y_\gamma} (u_{s_2 s_7}^m u_{s_7 s_4}^m u_{s_4 s_6}^m u_{s_6 s_2}^m)^{y_\gamma} u_{s_2 s_6}^m (u_{s_6 s_4}^m)^{y_\gamma}, \quad (3.15)$$

$$\begin{aligned} l_{n,\mu}^m(x_\beta, x_\gamma) &= [f_{n,\mu}^m(x_\gamma)^{-1}]^{y_\beta} [f_{n,\mu}^m(x_\gamma) k_{n,\mu}^m(x_\gamma) g_{n,\mu}^m(x_\gamma)^{-1} h_{n,\mu}^m(x_\gamma)^{-1}]^{y_\beta} \\ &\quad \cdot h_{n,\mu}^m(x_\gamma) [g_{n,\mu}^m(x_\gamma)]^{y_\beta}, \end{aligned} \quad (3.16)$$

$$S_{n,\mu}^m(x_\alpha, x_\beta, x_\gamma) = (u_{s_0 s_3}^m)^{y_\gamma} [f_{n,\mu}^m(x_\gamma)]^{y_\beta} [l_{n,\mu}^m(x_\beta, x_\gamma)]^{y_\alpha}. \quad (3.17)$$

Here, $y_\lambda \equiv x_\lambda - n_\lambda$ and $0 \leq y_\lambda \leq 1$ for $\lambda = \alpha, \beta, \gamma$. We have constructed $S_{n,\mu}^m(x_\alpha, x_\beta, x_\gamma)$ based on the link variable $u(n, \mu)$; we simply write it as $S_{n,\mu}^m(u; x) = S_{n,\mu}^m(x_\alpha, x_\beta, x_\gamma)[u]$ in what follows.

As a result, we have the transition function at $\forall x \in f(n, \mu)$ as

$$g_{n,\mu}(u; x) \equiv S_{n,\mu}^{n-\hat{\mu}}(u; x)^{-1} g_{n,\mu}(u; n) S_{n,\mu}^n(u; x), \quad g_{n,\mu}(u; n) = w^{n-\hat{\mu}}(u; n) w^n(u; n)^{-1}, \quad (3.18)$$

Then, we naively define $v_{n,\mu}(x) = g_{n,\mu}(U; x)$ and $\check{v}_{n,\mu}(x) = g_{n,\mu}(\check{U}; x)$. $v_{n,\mu}(x)$ obeys the twisted boundary condition,

$$v_{n,\mu}(x + L\hat{\nu}) = g_\nu(n - \hat{\mu})^{-1} v_{n,\mu}(x) g_\mu(n), \quad (3.19)$$

while $\check{v}_{n,\mu}(x)$ is periodic.

3.2.2 Admissibility condition

To obtain the well-defined interpolation $S_{n,\mu}^m(U; x)$, we should impose an *admissibility* condition. Here, for simplicity, we consider the $SU(2)$ gauge theory.⁶ First note that, to make the lattice action density small, the plaquette P is in a neighborhood of 1, and $\check{P}$ is $e^{\frac{2\pi i}{N} z_{\mu\nu}}$ or 1. $S_{n,\mu}^m(U; x)$ is a function with respect to the plaquettes P , and has a structure such as P^y with $0 \leq y \leq 1$ (see Sect. 3.2.1). Then, $P = -1$ is ill-defined and so such configurations are called exceptional. Supposing all combinations in $S_{n,\mu}^m(U; x)$ are well-defined, we have the admissibility condition, $\text{tr}[1 - P] < \varepsilon$. In Ref. [3], it is proved that there exists $\varepsilon > 0$ for $\forall N$. On the other hand, for $S_{n,\mu}^m(\check{U}; x)$, the situation is more complicated because we cannot easily choose a branch of $(\check{P})^y \sim (e^{\frac{2\pi i}{N} z_{\mu\nu}})^y$. To this end, let us construct $S_{n,\mu}^m(\check{U}; x)$ from $S_{n,\mu}^m(U; x)$. That is, since P can be rewritten as $e^{-\frac{2\pi i}{N} z_{\mu\nu}} \check{P}$, we can define

$$\left(e^{-\frac{2\pi i}{N} z_{\mu\nu}} \check{P} \right)^y = e^{-\frac{2\pi i}{N} (z_{\mu\nu} + NM_{\mu\nu})y} \left(e^{2\pi i M_{\mu\nu}} \check{P} \right)^y, \quad (3.20)$$

where $M_{\mu\nu}$ is an integer such that

$$\begin{cases} 0 \leq z_{\mu\nu} + NM_{\mu\nu} < N & \text{for } \mu < \nu, \\ z_{\mu\nu} + NM_{\mu\nu} = -z_{\nu\mu} - NM_{\nu\mu} & \text{for } \mu > \nu. \end{cases} \quad (3.21)$$

Also for a product of k plaquettes,

$$\left(\prod_{\ell=1}^k e^{-\frac{2\pi i}{N} z_\ell} \check{P}_\ell \right)^y = e^{-\frac{2\pi i}{N} \sum_{\ell=1}^k (z_\ell + NM_\ell)y} \left(e^{2\pi i \sum_{\ell=1}^k M_\ell} \prod_{\ell=1}^k \check{P}_\ell \right)^y, \quad (3.22)$$

where $z_\ell = z_{\mu_\ell \nu_\ell}$ with Eq. (3.21). Note that $|\sum_\ell (z_\ell + NM_\ell)| < Nk$. In what follows, we redefine $z_{\mu\nu} + NM_{\mu\nu}$ as $z_{\mu\nu}$ so that $0 \leq z_{\mu\nu} < N$ for $\mu < \nu$. Again, $\check{P} = e^{\frac{2\pi i}{N} z_{\mu\nu}} \times (-1)$ is

⁶ We can generalize discussions below to any compact gauge group. Admissible configurations should be close to those at the minimum in the classical continuum limit. Thus, such configurations are topologically on a disk. Since $SU(2)$ is topologically a sphere, removing simply one point on it, we have a desired admissibility.

not defined, and thus we have the same admissibility condition given by

$$\text{tr} \left[1 - e^{-\frac{2\pi i}{N} z_{\mu\nu} \check{P}} \right] < \varepsilon. \quad (3.23)$$

We should mention that $\check{v}_{n,\mu}(x)$ is an element of $U(N)$. This is because $(\check{P})^y \sim e^{\frac{2\pi i}{N} z_{\mu\nu} y} \in U(1)$. On lattice sites, that is, at $y = 0$ or 1 , this factor becomes $\mathbb{Z}_N$ so $\check{v}_{n,\mu} \in SU(N)$. When we write $v_{n,\mu}(x)$ in terms of $\check{v}_{n,\mu}(x)$, the extra factor of $z_{\mu\nu}$ in $v_{n,\mu}(x)$ appears from Eqs. (3.20) and (3.22), which is similar to $\varsigma_\mu(x)$ in the continuum theory. This factor, $e^{-\frac{2\pi i}{N} z_{\mu\nu} y}$, is also an element of $U(1)$. These $U(1)$ factors, $(\check{P})^y$ and $e^{-\frac{2\pi i}{N} z_{\mu\nu} y}$, cancel out by construction, that is, $1 \sim e^{-\frac{2\pi i}{N} z_{\mu\nu} y} (\check{P})^y = P^y \in SU(N)$; so $v_{n,\mu}(x) \in SU(N)$ is kept intact.

3.2.3 Fractional topological charge

Following the above construction, we can rewrite the transition function $v_{n,\mu}(x)$ in terms of a $z_{\mu\nu}$ dependent factor and $\check{v}_{n,\mu}(x)$. We obtain the transition function by, for $x \in f(n, \mu)$,

$$v_{n,\mu}(x) = \begin{cases} \omega_{n,\mu}(x) \check{v}_{n,\mu}(x) g_\mu(n - \hat{\mu}) & \text{for } n_\mu = L, \\ \omega_{n,\mu}(x) \check{v}_{n,\mu}(x) & \text{otherwise,} \end{cases} \quad (3.24)$$

where the loop factor $\omega_{n,\mu}(x)$ is defined by

$$\omega_{n,\mu}(x) \equiv \begin{cases} \exp \left(\frac{2\pi i}{N} \sum_{\nu > \mu} z_{\mu\nu} y_\nu \delta_{n_\nu, L-1} \right) & \text{for } x_\mu = 0 \bmod L, \\ 1 & \text{otherwise.} \end{cases} \quad (3.25)$$

The transition functions provide the cocycle condition given by, at $x \in p(n, \mu, \nu)$,

$$v_{n-\hat{\nu},\mu}(x) v_{n,\nu}(x) v_{n,\mu}(x)^{-1} v_{n-\hat{\mu},\nu}(x)^{-1} = \check{v}_{n-\hat{\nu},\mu}(x) \check{v}_{n,\nu}(x) \check{v}_{n,\mu}(x)^{-1} \check{v}_{n-\hat{\mu},\nu}(x)^{-1} = 1, \quad (3.26)$$

$$\omega_{n-\hat{\nu},\mu}(x) \omega_{n,\nu}(x) \omega_{n,\mu}(x)^{-1} \omega_{n-\hat{\mu},\nu}(x)^{-1} = \begin{cases} \exp \left(\frac{2\pi i}{N} z_{\mu\nu} \right) & x_\mu = x_\nu = 0 \bmod L, \\ 1 & \text{otherwise.} \end{cases} \quad (3.27)$$

Note that $v_{n,\mu}(x)$ is nontrivial because of the $\mathbb{Z}_N$ twisted boundary condition of it (3.19) though its cocycle condition is not relaxed by $\mathbb{Z}_N$. If one use $\tilde{v}_{n,\mu}(x) = \omega_{n,\mu}(x) \check{v}_{n,\mu}(x)$ obeying the periodic boundary condition, we find that $\tilde{v}$ satisfies the same cocycle condition as ω (3.27).

Substituting $v_{n,\mu}(x)$ into the topological charge $\mathcal{Q}$ [56, 57]

$$\begin{aligned} \mathcal{Q} = & -\frac{1}{24\pi^2} \sum_{n \in \Gamma} \sum_{\mu, \nu, \rho, \sigma} \epsilon_{\mu\nu\rho\sigma} \left\{ 3 \int_{p(n, \mu, \nu)} d^2x \text{tr} \left[(v_{n,\mu} \partial_\rho v_{n,\mu}^{-1}) (v_{n-\hat{\mu},\nu}^{-1} \partial_\sigma v_{n-\hat{\mu},\nu}) \right] \right. \\ & \left. + \int_{f(n, \mu)} d^3x \text{tr} \left[(v_{n,\mu}^{-1} \partial_\nu v_{n,\mu}) (v_{n,\mu}^{-1} \partial_\rho v_{n,\mu}) (v_{n,\mu}^{-1} \partial_\sigma v_{n,\mu}) \right] \right\}, \end{aligned} \quad (3.28)$$

we have

$$\mathcal{Q} = -\frac{1}{8N} \sum_{\mu, \nu, \rho, \sigma} \epsilon_{\mu\nu\rho\sigma} z_{\mu\nu} z_{\rho\sigma} + \check{\mathcal{Q}}, \quad (3.29)$$

where $\check{\mathcal{Q}}$ is the topological charge with respect to the periodic variable $\check{U}$, and we have used the similar identity to Eq. (2.12) for the cross terms. The first term is a fractional part with $\frac{1}{N}\mathbb{Z}$, and one finds $\check{\mathcal{Q}} \in \mathbb{Z}$ thanks to the cocycle condition (3.26).

3.3 Remarks on the relation with other constructions

We make remarks on other constructions based on different ideas from ours: $v_{n,\mu}(x)$ is similar to the transition function $\tilde{v}_{n,\mu}(x)$ defined in Ref. [3] when we consider a specific 't Hooft flux at the corner. The different point is that $\tilde{v}_{n,\mu}(x)$ is periodic. Thus, $v_{n,\mu}(x) = \tilde{v}_{n,\mu}(x) g_\mu(n - \hat{\mu})$ for $n_\mu = L$.

Our definition of $\check{v}_{n,\mu}(x)$ cannot be realized as $g_{n,\mu}(\check{U}; x)$ any longer without any other information. Actually, it is not necessary to be based on the underlying periodic variable $\check{U}(n, \mu)$. As an alternative way, we can define $\check{v}_{n,\mu}(x)$ as $v_{n,\mu}(x)$ divided by the loop factor at the minimum, as defined in the continuum by $\varsigma_\mu(x)$ in Sect. 2. The discussions in the paper can be also considered in this sense.

We ask how the definition of $\check{v}_{n,\mu}(x)$ is related to that defined for the $U(1)$ lattice gauge theory in Ref. [1]. The idea is, as an other prescription, to replace the plaquette by it to the N th power; that is, $\check{P}^y \rightarrow (P^N)^{y/N}$ to impose the $\mathbb{Z}_N$ 1-form gauge symmetry as we will define later. This simple prescription removes the branch ambiguity above, but is quite subtle and in fact not rigorous. We first recognize that, because $(e^{-\frac{2\pi i}{N} z_{\mu\nu}} \check{P})^N = \check{P}^N$, we lose any information on the $\mathbb{Z}_N$ center, and then $\check{v}_{n,\mu}(x) = v_{n,\mu}(x)$ with the “stronger” admissibility ε/N . On the contrary, it looks like $e^{-\frac{2\pi i}{N} z_{\mu\nu} y} (\check{P}^N)^{y/N}$, and then, the information on the branch would be guessed from the first factor thanks to the $\mathbb{Z}_N$ 1-form gauge invariance. Therefore, though this prescription is not a sure way, it could hold up on a robust principle of the 1-form gauge symmetry.

4 1-form gauge invariance on the lattice

4.1 Gauging the $\mathbb{Z}_N$ 1-form symmetry on the lattice

We have started with the similar procedure to that in the continuum by van Baal, and obtained the fractional topological charge with the 't Hooft flux at the corner of Γ . Actually, it is known that the structure of the $SU(N)/\mathbb{Z}_N$ principal bundle from lattice theories is much more general and robust, whose most important principle is based on the locality,

$SU(N)$ gauge invariance, and $\mathbb{Z}_N$ 1-form gauge invariance. In this section, let us consider the $\mathbb{Z}_N$ 1-form gauge invariance [16, 24, 54], which plays a crucial role in this robustness [3].

From the constant flux $z_{\mu\nu}$, we introduce a lattice field $z_{\mu\nu}(n)$, as a $\mathbb{Z}_N$ 2-form gauge field,

$$z_{\mu\nu}(n) \equiv z_{\mu\nu} \delta_{n_\mu, L-1} \delta_{n_\nu, L-1}. \quad (4.1)$$

The loop factor is then given by

$$\omega_{n,\mu}(x) = \exp \left[\frac{2\pi i}{N} \sum_{\nu > \mu} z_{\mu\nu}(n - \hat{\mu}) y_\nu \right], \quad (4.2)$$

and the twisted cocycle condition is

$$v_{n-\hat{\nu},\mu}(x) v_{n,\nu}(x) v_{n,\mu}(x)^{-1} v_{n-\hat{\mu},\nu}(x)^{-1} = \exp \left[\frac{2\pi i}{N} z_{\mu\nu}(n - \hat{\mu} - \hat{\nu}) \right]. \quad (4.3)$$

We give the $\mathbb{Z}_N$ 1-form gauge transformation:

$$\check{U}(n, \mu) \mapsto \exp \left[\frac{2\pi i}{N} z_\mu(n) \right] \check{U}(n, \mu) \quad z_\mu(n) \in \mathbb{Z}, 0 \leq z_\mu(n) < N. \quad (4.4)$$

Moreover, we assume

$$z_{\mu\nu}(n) \mapsto z_{\mu\nu}(n) + \Delta_\mu z_\nu(n) - \Delta_\nu z_\mu(n) + N M_{\mu\nu}(n), \quad (4.5)$$

where we have defined the forward difference, $\Delta_\mu f(n) \equiv f(n + \hat{\mu}) - f(n)$, and $M_{\mu\nu}(n)$ is required to restrict the 2-form gauge field $z_{\mu\nu}(n)$ to

$$\begin{cases} 0 \leq z_{\mu\nu}(n) < N & \text{for } \mu < \nu, \\ z_{\mu\nu}(n) = -z_{\nu\mu}(n) & \text{for } \mu > \nu. \end{cases} \quad (4.6)$$

Under this 1-form transformation, $\check{P} \mapsto e^{-\frac{2\pi i}{N}(\Delta_\mu z_\nu - \Delta_\nu z_\mu + N M_{\mu\nu})} \check{P}$ by our choice of the branch.

As shown in Appendix A of Ref. [1] (see also Ref. [81]), the consistency of transition functions among the intersection of eight hypercubes leads us to the flatness of the $\mathbb{Z}_N$ 2-form gauge field $z_{\mu\nu}(n)$. We can see that $z_{\mu\nu}(n)$ satisfies the modulo N flatness condition,

$$\frac{1}{2} \sum_{\nu, \rho, \sigma} \epsilon_{\mu\nu\rho\sigma} \Delta_\nu z_{\rho\sigma}(n) = 0 \bmod N. \quad (4.7)$$

Therefore, we find that, under the $\mathbb{Z}_N$ 1-form gauge transformation,

$$\begin{aligned} \check{v}_{n,\mu}(x) \mapsto & \exp \left\{ -\frac{2\pi i}{N} \sum_{\nu > \mu} [\Delta_\mu z_\nu(n - \hat{\mu}) - \Delta_\nu z_\mu(n - \hat{\mu}) + NM_{\mu\nu}(n - \hat{\mu})] y_\nu \right\} \\ & \times \exp \left[\frac{2\pi i}{N} z_\mu(n - \hat{\mu}) \right] \check{v}_{n,\mu}(x). \end{aligned} \quad (4.8)$$

The first factor can be canceled against the transformation of $\omega_{n,\mu}(x)$, which depends on the gauge field $z_{\mu\nu}(n)$, so we have

$$v_{n,\mu}(x) \mapsto \exp \left[\frac{2\pi i}{N} z_\mu(n - \hat{\mu}) \right] v_{n,\mu}(x). \quad (4.9)$$

The fractional topological charge $\mathcal{Q}$

$$\mathcal{Q} = -\frac{1}{8N} \sum_{n \in \Gamma} \sum_{\mu, \nu, \rho, \sigma} \epsilon_{\mu\nu\rho\sigma} z_{\mu\nu}(n) z_{\rho\sigma}(n + \hat{\mu} + \hat{\nu}) + \check{\mathcal{Q}} \quad (4.10)$$

is invariant under the $\mathbb{Z}_N$ 1-form gauge transformation, while $\check{\mathcal{Q}}$ is not. The shift of $\check{\mathcal{Q}}$ should vanish owing to the shift of the first term (see Appendix A in Ref. [1]).

4.2 Reproducing the Kapustin–Seiberg prescription

To compare some properties between the lattice and continuum theories [54], one may construct the $SU(N)/\mathbb{Z}_N$ principal bundle from the lattice $U(N)$ gauge theory. Starting from the above construction by the lattice $SU(N)$ gauge theory, we perform the $U(1)$ 1-form transformation given by

$$\check{U}(n, \mu) \mapsto \exp[i\lambda_\mu(n)] \check{U}(n, \mu) \in U(N) \quad \lambda_\mu(n) \in \mathbb{R}, \quad 0 \leq \lambda_\mu(n) < 2\pi. \quad (4.11)$$

Now we define the $U(1)$ 2-form gauge field

$$\lambda_{\mu\nu}(n) \equiv \frac{2\pi}{N} z_{\mu\nu}(n) + \Delta_\mu \lambda_\nu(n) - \Delta_\nu \lambda_\mu(n) + 2\pi K_{\mu\nu}(n), \quad (4.12)$$

where $K_{\mu\nu}(n) \in \mathbb{Z}$ is required to restrict $\lambda_{\mu\nu}(n)$ to

$$\begin{cases} 0 \leq \lambda_{\mu\nu}(n) < 2\pi & \text{for } \mu < \nu, \\ \lambda_{\mu\nu}(n) = -\lambda_{\nu\mu}(n) & \text{for } \mu > \nu. \end{cases} \quad (4.13)$$

After successive $U(1)$ 1-form transformations, a generic $\lambda_{\mu\nu}(n)$ transforms as

$$\lambda_{\mu\nu}(n) \mapsto \lambda_{\mu\nu}(n) + \Delta_\mu \lambda_\nu(n) - \Delta_\nu \lambda_\mu(n) + 2\pi K_{\mu\nu}(n). \quad (4.14)$$

Then, the transition function $v_{n,\mu}(x) \in U(N)$ transforms as

$$v_{n,\mu}(x) \mapsto \exp[i\lambda_\mu(n - \hat{\mu})] v_{n,\mu}(x), \quad (4.15)$$

and satisfies the cocycle condition

$$v_{n-\hat{\nu},\mu}(x)v_{n,\nu}(x)v_{n,\mu}(x)^{-1}v_{n-\hat{\mu},\nu}(x)^{-1} = \exp[i\lambda_{\mu\nu}(n - \hat{\mu} - \hat{\nu})]. \quad (4.16)$$

We have the topological charge

$$\mathcal{Q} = -\frac{N}{32\pi^2} \sum_{n \in \Gamma} \sum_{\mu, \nu, \rho, \sigma} \epsilon_{\mu\nu\rho\sigma} \lambda_{\mu\nu}(n) \lambda_{\rho\sigma}(n + \hat{\mu} + \hat{\nu}) + \check{\mathcal{Q}}. \quad (4.17)$$

This is, in fact, $U(1)$ 1-form invariant, and thus, takes the same fractional value as before. This means that we can regard $\lambda_{\mu\nu}(n)$ as a field strength in the lattice $U(1)/\mathbb{Z}_N$ gauge theory; as already shown in Ref. [1], one can show that the first term in $\mathcal{Q}$ is $N \times \frac{1}{N^2} \mathbb{Z}$.

The field strength is defined by

$$F_{\mu\nu}(n) \equiv \frac{1}{i} \ln [P(n, \mu, \nu)], \quad -\pi < F_{\mu\nu}(n) \leq \pi \quad (4.18)$$

and $\check{F}_{\mu\nu}(n)$ by $\check{P}(n, \mu, \nu)$, in the range (or branch) that those satisfy the admissibility condition (3.23). From $P(n, \mu, \nu) = e^{-i\lambda_{\mu\nu}(n)} \check{P}(n, \mu, \nu) \in SU(N)$, $F_{\mu\nu} = \check{F}_{\mu\nu} - \lambda_{\mu\nu} \in \mathfrak{su}(N)$ and $\check{F}_{\mu\nu} \in \mathfrak{u}(N)$; we have

$$\text{tr } \check{F}_{\mu\nu}(n) = N\lambda_{\mu\nu}(n). \quad (4.19)$$

Also under the $U(1)$ 1-form gauge transformation, the field strength $F_{\mu\nu}(n)$ is invariant, and $\check{F}_{\mu\nu}(n)$ transforms as

$$\check{F}_{\mu\nu}(n) \mapsto \check{F}_{\mu\nu}(n) + \Delta_\mu \lambda_\nu(n) - \Delta_\nu \lambda_\mu(n) + 2\pi K_{\mu\nu}(n). \quad (4.20)$$

These expressions provide good agreement with those in the continuum theory by the Kapustin–Seiberg prescription [54]. Therefore, our fully-regularized construction can naturally reproduce non-Abelian gauge theories with discrete higher-form symmetries in the continuum limit.

We make some comments here: First, in the more sophisticated and minimal construction given in Ref. [3], we can define the 1-form invariant $F_{\mu\nu}(n)$ only, and so the relation with the coupling with higher-form gauge fields in the continuum is not obvious. Also, if we use the $\mathbb{Z}_N$ 2-form gauge field $z_{\mu\nu}(n)$, its continuum limit seems to be puzzling because of no *smooth and integer* fields in the continuum description. This difficulty reminds us that there is a certain procedure required for such a coupling with a $\mathbb{Z}_N$ field in the continuum, which

would be described by a pair of $U(1)$ gauge fields. We can describe it by the lattice gauge field $\lambda_{\mu\nu}(n)$.

On the other hand, lattice gauge theories can be coupled quite simply with lattice integer fields, while there are subtleties in the continuum theory since cohomological operations are needed such as the Pontryagin square [24, 54]. The Kapustin–Seiberg prescription gives a simplified explanation for discrete gauge fields by embedding it to $U(1)$. Thus, we see the de Rham cohomology but not the Čech cohomology; to circumvent this issue, the wedge product in expressions as a result should be replaced by the Pontryagin square. Such issues make some studies in continuum apparently difficult.

Recently, the definition of (higher-)cup products, from which the Pontryagin square can be constructed, on the hypercubic lattice is given in Ref. [82].⁷ Then, in terms of the cochain and the cup product on the hypercubic lattice, Eqs. (4.10) and (4.17) can be written as

$$\mathcal{Q} = -\frac{1}{2N} \sum_{\text{hypercube} \in \Gamma} z \cup z + \check{\mathcal{Q}} = -\frac{N}{8\pi^2} \sum_{\text{hypercube} \in \Gamma} \lambda \cup \lambda + \check{\mathcal{Q}}. \quad (4.21)$$

5 Applications to higher-group and non-invertible symmetries

5.1 Higher-group structure in modified $SU(N)$ gauge theory

As a simple application, let us consider the higher-group structure in the $SU(N)$ gauge theory modified as follows: The insertion of the delta function in the path integral, $\delta(q(n) - pc(n))$, restricts the instanton number to integral multiples of $p \in \mathbb{Z}$ [63–66]. Here we have defined $\mathcal{Q} = \sum_{n \in \Gamma} q(n)$, and $c(n)$ is the 4-form field strength of a compact $U(1)$ 3-form gauge field and $\sum_{n \in \Gamma} c(n) \in \mathbb{Z}$. Thus, perturbation theory is not affected at all and the local nature is unchanged, while globally or topologically speaking this modification possesses a quite nontrivial structure, called the higher-group symmetry [62].⁸ This theory provides an application of our construction of the $SU(N)/\mathbb{Z}_N$ principal bundle.

Now, let us consider the constraint, $q(n) - pc(n) = 0$. At first sight, any nontrivial configuration of $z_{\mu\nu}(n)$ is forbidden, so gauging the $\mathbb{Z}_N$ 1-form symmetry seems to be impossible. Now, by introducing the $\mathbb{Z}_p$ 3-form symmetry and gauging these two symmetries simultaneously, we have a nontrivial theory with the 4-group structure. This can be immediately

⁷ For a torsion-free manifold, we can also use the integral lift [24, 54] as another possible choice.

⁸ In topological lattice gauge theories, we can also observe the higher-group structure; see Refs. [83, 84].

realized by the replacement as $c \rightarrow c - \frac{1}{Np}w$, where a 4-form field $w(n) \in \mathbb{Z}$; then

$$q(n) - pc(n) + \frac{1}{N}w(n) = 0. \quad (5.1)$$

Because all fractional contributions from $q(n)$ by $z_{\mu\nu}(n)$ can be absorbed into $w(n)$, one can obtain nontrivial configurations of $z_{\mu\nu}(n)$. Note that, by construction, we see the 3-form gauge symmetry,

$$w \mapsto w + \delta\omega^s \mod Np, \quad c \mapsto c + \frac{1}{Np}\delta\omega^s \mod 1 \quad (5.2)$$

Here we assume that $\omega_{\mu\nu\lambda}^s(n) \in \mathbb{Z}$ satisfies

$$\sum_{\text{hypercube} \in \Gamma} \delta\omega^s = \frac{1}{3!} \sum_{n \in \Gamma} \sum_{\mu, \nu, \rho, \sigma} \epsilon_{\mu\nu\rho\sigma} \Delta_\mu \omega_{\nu\rho\sigma}^s(n) \in Np\mathbb{Z} \quad (5.3)$$

and $0 \leq \omega_{\mu\nu\lambda}^s(n) < Np$ for $\mu < \nu < \lambda$. Note that w is not transformed by $\mathbb{Z}_N$ 1-form symmetry; the 3-form symmetry is not $\mathbb{Z}_p$ but $\mathbb{Z}_{Np}$ because the (strict) 4-group mixes the $\mathbb{Z}_N$ 1-form symmetry into the *physical* $\mathbb{Z}_p$ 3-form symmetry.

As another perspective, we can consider the use of $\Omega(n) \in \mathbb{R}$ such that $c \rightarrow c - \frac{1}{Np}\Omega$. Then, we have

$$q(n) - pc(n) + \frac{1}{N}\Omega(n) = 0, \quad (5.4)$$

where we assume that $\sum_{n \in \Gamma} \Omega(n) \in \mathbb{Z}$ to remove the fractionality in the first term. We now define $\tilde{\Omega}(n) \in \mathbb{R}$ by

$$\tilde{\Omega} \equiv \frac{1}{N}\Omega - \frac{1}{2N}z \cup z. \quad (5.5)$$

Note that $\sum_{\text{hypercube} \in \Gamma} \tilde{\Omega} \in \mathbb{Z}$. Therefore, by using $\tilde{\mathcal{Q}} = \sum_{n \in \Gamma} \tilde{q}(n) \in \mathbb{Z}$, we find that the constraint becomes

$$\tilde{q}(n) - pc(n) + \tilde{\Omega}(n) = 0. \quad (5.6)$$

From the definition of $\tilde{\Omega}(n)$ (5.5), $\tilde{\Omega}(n)$ is not invariant any longer under the $\mathbb{Z}_N$ 1-form gauge transformation, while $[\tilde{q}(n) + \tilde{\Omega}(n)]$ is invariant. One can find that, $\sum_{n \in \Gamma} \tilde{\Omega}(n) \in \mathbb{Z}$ holds after we perform the 1-form gauge transformation.

One can compel $\tilde{\Omega}(n) \in \mathbb{R}$ to become $\tilde{\Omega}(n) = \tilde{w}(n) \in \mathbb{Z}$. This is always possible since we can throw away the real or fractional part apart from integers in $\tilde{\Omega}(n)$ into the gauge redundancy of $c(n)$, by using the *continuum* 3-form transformation,

$$\tilde{\Omega} \mapsto \tilde{\Omega} + \delta\omega^w \mod p, \quad c \mapsto c + \frac{1}{p}\delta\omega^w \mod 1, \quad (5.7)$$

where $\omega_{\mu\nu\lambda}^w(n) \in \mathbb{R}$. The “ $\mathbb{Z}_N$ 1-form gauge transformation” is redefined as the original 1-form gauge transformation and such a 3-form transformation to be set to integers. This

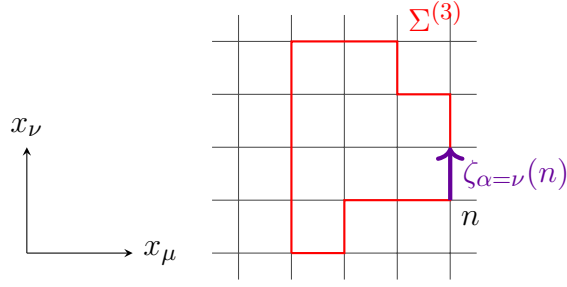


Fig. 3 Illustration of the definition of the configuration $\zeta_\alpha(n)$ (5.8). E.g., on the $\mu\nu$ plane, the arrow starting from n indicates the $\alpha = \nu$ direction tangent to the surface $\Sigma^{(3)}$, and another direction μ is vertical.

theory possesses the mixture of the $\mathbb{Z}_N$ 1-form and continuum 3-form gauge symmetries, and the *discrete* $\mathbb{Z}_p$ 3-form gauge symmetry as the weak structure of 4-group symmetry.

5.2 Non-invertible topological condensation defect among $SU(N)$

Let $G^{[p]}$ be a d -dimensional p -form symmetry. If $G^{[p]}$ can be gauged on a co-dimension- q surface with $q \leq p + 1$ (higher-gauging [38]), a topological defect on this surface arises. This is called the condensation defect. Note that the condensation defect can be invertible or non-invertible. Now, since the 4-dimensional $SU(N)$ gauge theory has the $\mathbb{Z}_N$ 1-form symmetry, the higher-gauging on a co-dimension-1 surface is nontrivial. In this subsection, we consider the condensation defect induced by this.

Now, on a closed 3-dimensional surface $\Sigma^{(3)} \subset \Gamma$, the higher-gauging of $z_{\mu\nu}(n)$ is given by configurations so that

$$z_{\mu\nu}(n) = \zeta_\alpha(n) \delta(\Sigma^{(3)}), \quad (5.8)$$

where the Kronecker delta denotes

$$\delta(\Sigma) = \begin{cases} 1 & \text{on } \Sigma, \\ 0 & \text{otherwise.} \end{cases} \quad (5.9)$$

For two directions along μ and ν for $z_{\mu\nu}(n)$, the $\alpha \in \{\mu, \nu\}$ direction of $\zeta_\alpha(n)$ is tangent to the surface $\Sigma^{(3)}$, while the other direction is vertical to it (see Fig. 3); otherwise $\zeta_\alpha(n) = 0$.

We define the Wilson plaquette action with the *effectively* 1-form gauge field $\zeta_\alpha(n)$ by

$$\begin{aligned} S_W[U; \zeta] &\equiv \sum_n \sum_{\mu, \nu} \beta \{ \text{tr} [1 - P(n, \mu, \nu)] + \text{c.c.} \} \\ &= \sum_n \sum_{\mu, \nu} \beta \left\{ \text{tr} \left[1 - e^{-\frac{2\pi i}{N} z_{\mu\nu}(n)} \check{P}(n, \mu, \nu) \right] + \text{c.c.} \right\}. \end{aligned} \quad (5.10)$$

One can define the condensation defect $\mathcal{C}(\Sigma^{(3)})$ on the lattice by letting $\zeta_\alpha(n)$ be a dynamical field on $\Sigma^{(3)}$, that is,

$$\langle \mathcal{C}(\Sigma^{(3)}) \rangle \equiv \int \mathcal{D}U \int_{\Sigma^{(3)}} \mathcal{D}\zeta e^{-S_W[U; \zeta]}. \quad (5.11)$$

We consider a loop ℓ along S^1 on the closed surface $\Sigma^{(3)} = S^1 \times S^2$. Then, the Wilson loop along ℓ , W_ℓ , is the charged object associated to the $\mathbb{Z}_N$ gauge field $\zeta_\alpha(n)$. The expectation value with the condensation defect is given by

$$\langle W_\ell \mathcal{C}(\Sigma^{(3)}) \rangle \propto \left(\sum_k e^{\frac{2\pi i}{N} k} \right) \langle W_\ell \rangle = 0, \quad (5.12)$$

because the summation over all $\mathbb{Z}_N$ charges (i.e., the path integral over all $\mathbb{Z}_N$ sectors) vanishes. If this expectation value does not vanish, $\langle W_\ell \mathcal{C}(\Sigma^{(3)}) \rangle \neq 0$, there exists an inverse of $\mathcal{C}(\Sigma^{(3)})$. This implies that the condensation defect we have constructed is non-invertible.

6 Lattice realization of axial $U(1)$ non-invertible symmetry

In a usual sense, the quantum anomaly leads to loss of symmetry. However, the recent studies given in Refs. [40, 41] shows that, in a system with the ABJ anomaly, there exists the axial $U(1)$ non-invertible symmetry when the axial $U(1)$ transformation is restricted to rational rotation angles. In this section, we make attempts to reconstruct the axial $U(1)$ non-invertible symmetry on a completely lattice regularized framework.

6.1 Review of fractional quantum Hall effect and non-invertibility

In the continuum, a naive axial $U(1)$ current is not conserved due to the ABJ anomaly,

$$d \star j = \frac{1}{4\pi^2} F \wedge F, \quad (6.1)$$

and then a naive symmetry operator $U_\alpha(\mathcal{M}_3) = \exp\left(\frac{i\alpha}{2} \int_{\mathcal{M}_3} \star j\right)$ on a 3-dimensional closed surface $\mathcal{M}_3$ is not topological, where α is the axial $U(1)$ rotation angle. It is well-known that, by adding the Chern–Simons form, one can construct the modified axial $U(1)$ current

$$d \left(\star j - \frac{1}{4\pi^2} A \wedge dA \right) = 0, \quad (6.2)$$

to which a symmetry operator associated, $\hat{U}_\alpha(\mathcal{M}_3)$, becomes topological under deformations of $\mathcal{M}_3$. However, the Chern–Simons action on $\mathcal{M}_3$ with a coefficient α in $\hat{U}_\alpha(\mathcal{M}_3)$ is not gauge invariant since this corresponds to a *non-integer* level Chern–Simons theory. This modification cannot be regarded as a physical one [85].

Now, let us consider the following topological field theory, the fractional quantum Hall state [69]:

$$i \int_{\mathcal{M}_3} \left(\frac{N}{4\pi} a \wedge da + \frac{1}{2\pi} a \wedge dA \right). \quad (6.3)$$

Here, the first term denotes the level- N Chern–Simons term, and the second one is the coupling in the form $A_\mu J^\mu$ with a current J . Naively speaking, the equation of motion is given by $a = -A/N$, and then we may rewrite this system as a level- $(1/N)$ Chern–Simons theory. Of course, this is not correct because Eq. (6.3) is gauge invariant while the fractional level Chern–Simons term is not; the equation of motion holds only on each patch but not globally.

As I mentioned, the fractional quantum Hall state (6.3) is gauge invariant. Therefore, we can rewrite the gauge non-invariant operator $\hat{U}_\alpha(\mathcal{M}_3)$ as a gauge-invariant one with a fractional rotation angle $2\pi/N$,

$$D_{1/N}(\mathcal{M}_3) = \exp \left[i \int_{\mathcal{M}_3} \left(\frac{\pi}{N} \star j + \frac{N}{4\pi} a \wedge da + \frac{1}{2\pi} a \wedge dA \right) \right]. \quad (6.4)$$

This is a desired operator so that $D_{1/N}$ is topological (i.e., symmetry operator) and gauge invariant (i.e., physical operator). On the other hand, one can find that $D_{1/N}$ is not unitary as

$$D_{1/N} \times D_{1/N}^\dagger = \mathcal{C} \neq 1, \quad (6.5)$$

where $\mathcal{C}$ is a condensation operator. This indicates that we have observed the *non-invertible* symmetry.

6.2 Towards lattice implementation of fractional quantum Hall state

To realize the above axial $U(1)$ non-invertible symmetry on the lattice must be hindered by the following hard problems:

- (1) Lattice formulation of (anomaly-free) chiral gauge theory,
- (2) Consistency in anomalous gauge theory,
- (3) Lattice formulation of the $U(1)$ Chern–Simons theory.

For the first issue, it is known, if the target theory is the $U(1)$ chiral gauge theory, that we can employ the lattice formulation by Lüscher [13]. The lattice fermion couples to the lattice $U(1)$ gauge field through the γ_5 coupling based on the Ginsparg–Wilson relation [71–78].

In a sense of (2), one may wonder what is a consistent viewpoint of gauge theory with anomalous matter contents. I do not have any answer for the question given in Ref. [12]:

In other words, do we have to know an anomaly-free fundamental theory very precisely to study dynamics of chiral gauge theories at an (low) energy scale of our concern?

From the discussion by Faddeev and Shatashvili in 1980s [86, 87], some people had tried to construct a correct quantization of anomalous theory. For example, in the Harada–Tsutsui formalism [9], one can define a Faddeev–Popov determinant as

$$\Delta_f[A] \int dg \delta(f[A^g]) = 1, \quad (6.6)$$

where we introduced the gauge-fixing condition such that if $f[A] = 0$ a solution of $f[A^g] = 0$ is unique; g denotes a gauge transformation. Then we carry out “gauge average” of the Wess–Zumino action, which is an integration over g under the invariant measure, and, as a consequence, we have the gauge-invariant effective action with non-local terms.

The lattice formulation of anomalous gauge theory, however, suffers from an obstruction. This is a kind of no-go theorem between single-valued-ness, gauge invariance, smoothness and locality of the construction of the chiral fermion measure on the lattice [11, 12]. It is unfortunately not consistent to attempt naively to Lüscher’s lattice formulation of the $U(1)$ chiral gauge theory.

Now let us consider the $U(1) \times U(1)'$ gauge theory. The first $U(1)$ factor is for a physical and dynamical gauge group, and the latter $U(1)'$ is for an external and non-dynamical gauge group. We introduce two $U(1)$ lattice gauge fields by link variables,

$$u(n, \mu) \in U(1), \quad U(n, \mu) \in U(1), \quad (6.7)$$

In what follows, we denote quantities for dynamical fields by lowercase letters, which are subject to the path integral, while quantities for external fields by capital letters. Also assume periodic boundary conditions hold:

$$u(n + L\hat{\nu}, \mu) = u(n, \mu), \quad U(n + L\hat{\nu}, \mu) = U(n, \mu). \quad (6.8)$$

Note that $u(n, \mu)$ couples to fermion fields in a vector-like way while the *external* $U(1)$ gauge field $U(n, \mu)$ couples to fermion fields in a chirally-asymmetric way through γ_5 (see later). The role of $U(n, \mu)$ is to manipulate the axial rotation angle, and then to define the topological defect, the axial $U(1)$ non-invertible symmetry. Actually, the $U(1)'$ sector possesses the gauge anomaly to reproduce the axial $U(1)$ anomaly, and non-local dependence in the fermion measure as the above no-go theorem; but we will find this is not problematic because $U(n, \mu)$ is not dynamical and moreover pure-gauge.

Finally, for the issue (3), that is, lattice formulation of the $U(1)$ Chern–Simons theory, the following ways of evading difficulties would arise:

- The use of the modified Villain formulation instead of the Wilson-type one,
- “Gauge average” such as the Harada–Tsutsui formalism given by Karasik [10],
- For rational rotation angles $\alpha = 2\pi p/N$ ($p, N \in \mathbb{Z}$), coupling to a topological field theory (TQFT), the 3-dimensional level- N BF theory.

Our lattice setup is applicable to the gauge average and/or coupling to the BF theory; we will construct these below and then observe the non-invertible symmetry $\tilde{U}_\alpha(\tilde{\mathcal{M}}_3)$. Here, $\tilde{\mathcal{M}}_3$ is a homologically trivial 3-dimensional defect on the dual lattice of Γ ; one can find that $U(n, \mu)$ is a lattice analogue of the Poincaré dual of $\tilde{\mathcal{M}}_3$. As we will see, there are some subtleties in the Karasik prescription in our lattice formulation. Some points may be partially resolved in the case of $\alpha = 2\pi p/N$ by using the BF theory; a lattice formulation of the BF theory makes fusion rules computable.

6.3 $U(1) \times U(1)'$ lattice gauge theory

6.3.1 Gauge sector in the lattice action

At first we consider the gauge sector; fermion fields are now supposed to be residing on sites of Γ . Assume that $U(n, \mu)$ is pure-gauge:

$$U(n, \mu) = \Lambda(x)^{-1} \Lambda(n + \hat{\mu}), \quad \Lambda(n) \in U(1), \quad (6.9)$$

where $\Lambda(n + L\hat{\mu}) = \Lambda(n)$. The expectation value is defined by the functional integral,

$$\langle \mathcal{O} \rangle = \frac{1}{\mathcal{Z}} \int \mathcal{D}u e^{-S_G} \langle \mathcal{O} \rangle_F, \quad \langle \mathcal{O} \rangle_F = w[m] \int \mathcal{D}\psi \mathcal{D}\bar{\psi} e^{-S_F} \mathcal{O}, \quad (6.10)$$

where the partition function $\mathcal{Z}$ is determined by requiring $\langle 1 \rangle = 1$, and $\mathcal{D}u$ is given by the Haar measure at each link. For smoothness, we adopt a gauge action S_G following Ref. [13],

$$S_G = \frac{1}{4g_0^2} \sum_{n \in \Gamma} \sum_{\mu, \nu} \mathcal{L}_{\mu\nu}(n) \quad (6.11)$$

where g_0 is the bare coupling and

$$\mathcal{L}_{\mu\nu}(n) = \begin{cases} [f_{\mu\nu}(n)]^2 \{1 - [f_{\mu\nu}(n)]^2/\epsilon^2\}^{-1} & \text{if } |f_{\mu\nu}(n)| < \epsilon, \\ \infty & \text{otherwise,} \end{cases} \quad (6.12)$$

for $0 < \epsilon < \pi/3$. The field strength of $u(n, \mu)$ is defined in the same way as Eq. (4.18). The gauge action (6.11) is designed to impose the admissibility in terms of the field strength [57,

88]

$$\sup_{n,\mu,\nu} |f_{\mu\nu}(n)| < \varepsilon. \quad (6.13)$$

Each of topological sectors is labeled by the magnetic fluxes [13],

$$m_{\mu\nu} = \frac{1}{2\pi} \sum_{s,t=0}^{L-1} f_{\mu\nu}(n + s\hat{\mu} + t\hat{\nu}), \quad (6.14)$$

for which the factor $w[m]$ in Eq. (6.10) parameterized the relative weight and phase. The admissibility condition also implies that the field strength $f_{\mu\nu}(n)$ satisfies the Bianchi identity,

$$\sum_{\nu,\rho,\sigma} \epsilon_{\mu\nu\rho\sigma} \Delta_\nu f_{\rho\sigma}(n) = 0. \quad (6.15)$$

$U(n, \mu)$, on the other hand, is pure-gauge and so $F_{\mu\nu}(n) = 0$.

6.3.2 Fermion sector in the lattice action

We consider a multiplet of two left-handed “Weyl” fermion fields, ψ_α ($\alpha = 1, 2$), each of which possesses $U(1)$ charges under the gauge group $U(1) \times U(1)'$:

$$(e_1, e'_1) = (+1, -1), \quad (e_2, e'_2) = (-1, -1). \quad (6.16)$$

Note that the coupling to $U(1)'$ is chiral while the coupling to $U(1)$ is vector-like (Dirac). It is then natural to define the forward and backward covariant difference operators on fermion fields

$$\begin{aligned} \nabla_\mu \psi(n) &= R[u(n, \mu)] R[U(n, \mu)] \psi(n + \hat{\mu}) - \psi(n), \\ \nabla_\mu^* \psi(n) &= \psi(n) - R[u(n - \hat{\mu}, \mu)]^{-1} R[U(n - \hat{\mu}, \mu)]^{-1} \psi(n - \hat{\mu}), \end{aligned} \quad (6.17)$$

where

$$R[u(n, \mu)]_{\alpha\beta} = \delta_{\alpha\beta} u(n, \mu)^{e_\alpha}, \quad R[U(n, \mu)]_{\alpha\beta} = \delta_{\alpha\beta} U(n, \mu)^{e'_\alpha}. \quad (6.18)$$

The Wilson Dirac operator, D_W , and the overlap Dirac operator, D [72, 75], are given by

$$D_W = \frac{1}{2} [\gamma_\mu (\nabla_\mu + \nabla_\mu^*) - \nabla_\mu \nabla_\mu^*], \quad D = 1 - A(A^\dagger A)^{-1/2}, \quad A = 1 - D_W \quad (6.19)$$

We can find that the Ginsparg–Wilson relation [71] is fulfilled

$$\gamma_5 D + D \gamma_5 = D \gamma_5 D \quad (6.20)$$

with the chiral matrix γ_5 , and the γ_5 -hermiticity, $D^\dagger = \gamma_5 D \gamma_5$.

Now, the Ginsparg–Wilson relation (6.20) enables us to define exact chiralities of a lattice fermion. Defining the modified chiral matrix by [77, 78],

$$\hat{\gamma}_5 = \gamma_5(1 - D), \quad (6.21)$$

which satisfies $(\hat{\gamma}_5)^\dagger = \hat{\gamma}_5$, $(\hat{\gamma}_5)^2 = 1$, and $D\hat{\gamma}_5 = -\gamma_5 D$, we can introduce chirality projection operators by

$$\hat{P}_\pm = \frac{1}{2}(1 \pm \hat{\gamma}_5), \quad P_\pm = \frac{1}{2}(1 \pm \gamma_5), \quad (6.22)$$

In terms of the left-handed *Weyl* fermion on the lattice,

$$\hat{P}_- \psi = \psi, \quad \bar{\psi} P_+ = \bar{\psi}, \quad (6.23)$$

the fermion action in Eq. (6.10) with lattice exact chiralities is given by

$$S_F = \sum_{n \in \Gamma} \bar{\psi}(n) D \psi(n) = \sum_{n \in \Gamma} \bar{\psi}(n) P_+ D \hat{P}_- \psi(n). \quad (6.24)$$

We can employ Grassmann-odd expansion coefficients in

$$\psi(n) = \sum_j v_j(n) c_j, \quad \bar{\psi}(n) = \sum_k \bar{c}_k \bar{v}_k(n), \quad (6.25)$$

where $v_j(n)$ and $\bar{v}_k(n)$ are some orthonormal basis vectors in the projected space

$$\hat{P}_- v_j = v_j, \quad \bar{v}_k P_- = \bar{v}_k, \quad (6.26)$$

with $(v_k, v_j) = \delta_{kj}$ and $(\bar{v}_k^\dagger, \bar{v}_j^\dagger) = \delta_{kj}$.

6.3.3 Construction of the fermion integration measure

The fermion integration measure is defined under Eq. (6.25) by

$$\mathcal{D}\psi \mathcal{D}\bar{\psi} = \prod_j dc_j \prod_k d\bar{c}_k \quad (6.27)$$

An crucial point given in Ref. [13] is that basis vectors v_j depend on the gauge field through $\hat{P}_-$, and thus cannot be uniquely determined. This is because under a variation δ of the gauge field, from Eq. (6.26), $\hat{P}_+ \delta v_j = \delta \hat{P}_- v_j$. Therefore, $\hat{P}_- \delta v_j$ is not determined by Eq. (6.26) and basis vectors v_j are ambiguous. The choice of the basis vectors requires the smoothness and locality of the fermion integration measure [13].

We note that, in terms of the expansion coefficients in Eq. (6.25), the fermion action is rewritten as

$$S_F = \sum_{k,j} \bar{c}_k M_{kj} c_j, \quad M_{kj} = \sum_{n \in \Gamma} \bar{v}_k(n) D v_j(n). \quad (6.28)$$

Under infinitesimal variations of the gauge fields as

$$\delta_\eta u(n, \mu) = i\eta_\mu(n) u(n, \mu), \quad \delta_\eta U(n, \mu) = i\eta'_\mu(n) U(n, \mu), \quad (6.29)$$

the variation of the partition function is given by⁹

$$\delta_\eta \ln \det M = \text{Tr} \left(\delta_\eta D \hat{P}_- D^{-1} P_+ \right) - i\mathfrak{L}_\eta \quad (6.30)$$

with the measure term and measure currents [13],

$$\mathfrak{L}_\eta = i \sum_j (v_j, \delta_\eta v_j) \equiv \sum_{n \in \Gamma} \sum_\mu [\eta_\mu(n) j_\mu(n) + \eta'_\mu(n) J_\mu(n)]. \quad (6.31)$$

Now, the representation matrices being introduced by

$$t_{\alpha\beta} = \delta_{\alpha\beta} e_\alpha, \quad T_{\alpha\beta} = \delta_{\alpha\beta} e'_\alpha, \quad (6.32)$$

for the anomaly-free $U(1)$ chiral gauge theory, then we know the following:

Theorem. *There exists a smooth fermion integration measure such that the associated current coincides j_μ if j_μ is a given current as follows:*

- *The current depends smoothly and locally on all admissible link variables.*
- *The current is gauge invariant and, under lattice symmetries, transforms as an axial vector current.*
- *The functional $\mathfrak{L}_\eta = \sum_{n,\mu} \eta_\mu(n) j_\mu(n)$ satisfies the integrability condition*

$$\delta_\eta \mathfrak{L}_\zeta - \delta_\zeta \mathfrak{L}_\eta = i \text{Tr} \left(\hat{P}_- [\delta_\eta \hat{P}_-, \delta_\zeta \hat{P}_-] \right). \quad (6.33)$$

- *The current satisfies the “anomalous” conservation law*

$$\Delta_\mu^* j_\mu(n) = -\frac{1}{2} \text{tr} [\gamma_5 t D(x, y)]_{y \rightarrow x=n}, \quad (6.34)$$

where Δ^* denotes the backward difference.

To consider the $U(1) \times U(1)'$ gauge theory, which is in interest, the currents j and J must depend non-locally on the external gauge field $U(n, \mu)$. This non-locality in the fermion

⁹ Actually, there are some subtleties in the discussion here. This arises from the difference of the numbers of basis vectors, v_j and $\bar{v}_k$, some crucial effects of the finite lattice size L , and so on. See Refs. [8, 13].

integration measure is not physical because $U(n, \mu)$ is not dynamical. Also, the anomalous conservation laws should become

$$\Delta_\mu^* j_\mu = -\frac{1}{2} \text{tr} [\gamma_5 t D], \quad \Delta_\mu^* J_\mu = -\frac{1}{2} \text{tr} [\gamma_5 T D] + \frac{1}{2} \gamma (f \cup f), \quad (6.35)$$

with $\gamma = -1/(32\pi^2)$ and $\epsilon < 1/30$. We can prove this in an almost same way as in Ref. [13] by appropriately modifying the Theorem. In Appendix A, we elucidate the issue of the local counterterm which provides a desired form of the gauge anomaly.¹⁰

Finally, under the infinitesimal gauge transformation,

$$\eta_\mu(n) = -\Delta_\mu \omega(n), \quad \eta'_\mu(n) = -\Delta_\mu \Omega(n), \quad (6.36)$$

we obtain

$$\delta_\eta \langle \mathcal{O} \rangle_F = \langle \delta_\eta \mathcal{O} \rangle_F - \frac{i}{2} \gamma \sum_{\text{hypercube} \in \Gamma} (\Omega \cup f \cup f) \langle \mathcal{O} \rangle_F. \quad (6.37)$$

In the expectation value (6.37), gauge transformations on the fermion fields are given by

$$\delta_\eta \psi(n) = [i\omega(n)t + i\Omega(n)T] \psi(n), \quad \delta_\eta \bar{\psi}(n) = \bar{\psi}(n) [-i\omega(n)t - i\Omega(n)T]. \quad (6.38)$$

Equation (6.37) shows that the first $U(1)$ has no gauge anomaly while the second $U(1)'$ possesses the gauge anomaly with a definite structure. The anomaly relation (6.37) is the most fundamental relation for our construction of the symmetry operator in the next subsection.

6.4 Topological symmetry defect for the axial $U(1)$ non-invertible symmetry

6.4.1 Introduction of the topological defect

Let us first construct a defect which corresponds to the axial $U(1)$ (anomalous) symmetry on the lattice. We consider a co-dimension 1 (3-dimensional) oriented closed surface $\widetilde{\mathcal{M}}_3$ on the dual lattice of Γ in Fig. 4. Here, we assume that $\widetilde{\mathcal{M}}_3$ is homologically trivial so that $\widetilde{\mathcal{M}}_3 = \partial(\mathcal{V}_4)$ for a 4-dimensional region, $\exists \mathcal{V}_4$ on Γ .

A particular configuration of $U(n, \mu)$ for a defect is then given by

$$U(n, \mu) = \begin{cases} e^{+i\alpha/2} & \text{if the link } \langle n \rightarrow n + \hat{\mu} \rangle \text{ intersects } \widetilde{\mathcal{M}}_3 \text{ in positive direction,} \\ e^{-i\alpha/2} & \text{if the link } \langle n \rightarrow n + \hat{\mu} \rangle \text{ intersects } \widetilde{\mathcal{M}}_3 \text{ in negative direction,} \\ 1 & \text{otherwise,} \end{cases} \quad (6.39)$$

where $\alpha \in [0, 2\pi)$ is a fixed real number. Equation (6.39) defines a lattice analogue of the Poincaré dual of a homologically trivial surface $\widetilde{\mathcal{M}}_3$.

¹⁰ Naive computations lead us to the “anomalous” conservation law for j_μ even in the continuum theory. The information obtained in the Appendix is utilized in the construction of our lattice fermion here.

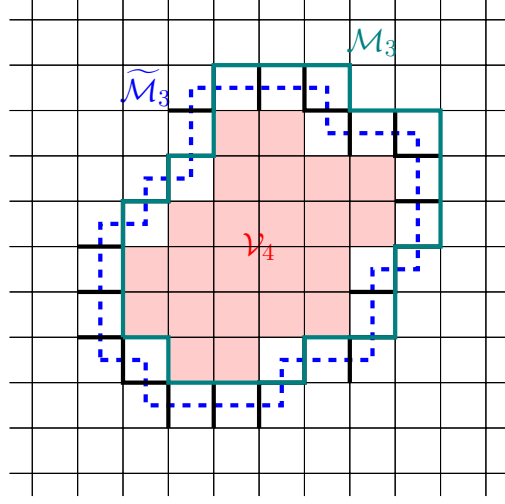


Fig. 4 Illustration of the 3-dimensional defect $\widetilde{\mathcal{M}}_3$ (the blue broken line) and the interior region, $\mathcal{V}_4$ (the red shaded area), and the corresponding 3-surface $\mathcal{M}_3$ (the green line) defining the boundary variables. This is a 2-dimensional slice of those. On the links indicated by thick ticks, each of $U(n, \mu)$ acquires a non-trivial phase $e^{\pm i\alpha/2}$ in Eq. (6.39).

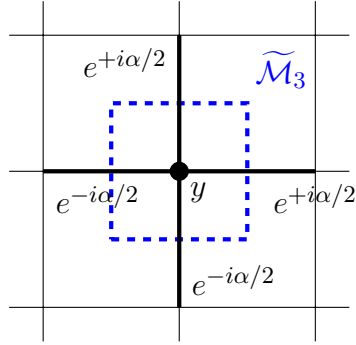


Fig. 5 A defect $\widetilde{\mathcal{M}}_3$ (the broken line) surrounding a single site y . The phases $e^{\pm i\alpha/2}$ on links are the values of $U(n, \mu)$ assigned according to the rule (6.39).

Here, we can deform freely the defect by a gauge transformation on $U(n, \mu)$, that is,

$$\Lambda(n) = e^{-i\frac{\alpha}{2}\delta_{ny}}, \quad (6.40)$$

eliminates the defect so that all $U(n, \mu)$ becomes unity with appropriate sites y . The path integral over the fermion fields in the previous subsection is, however, not invariant under the $U(1)'$ gauge transformation (6.40); we must have an anomaly. Now, suppose that $\widetilde{\mathcal{M}}_3$ surrounds a particular single site $y \in \Gamma$ as Fig. 5. From the *infinitesimal* gauge transformations

parameterized by $\delta\alpha$, $\Omega(n) = -\frac{1}{2}\delta\alpha\delta_{ny}$, Eq. (6.37) yields

$$\delta_\eta \langle \mathcal{O} \rangle_F = \langle \delta_\eta \mathcal{O} \rangle_F + \frac{i}{4} \delta\alpha \gamma(f \cup f)|_y \langle \mathcal{O} \rangle_F, \quad (6.41)$$

where the fermion fields inside the expectation value (see Eq. (6.38)) transform

$$\delta_\eta \psi(n) = \frac{i}{2} \delta\alpha \psi(n) \delta_{ny}, \quad \delta_\eta \bar{\psi}(n) = -\frac{i}{2} \delta\alpha \bar{\psi}(n) \delta_{ny}. \quad (6.42)$$

Thus, the finite gauge transformations with α promote the expectation value $\langle \mathcal{O} \rangle_F^{\widetilde{\mathcal{M}}_3}$ with the defect $\widetilde{\mathcal{M}}_3$ surrounding a single site y (Fig. 5) as

$$\langle \mathcal{O} \rangle_F^{\widetilde{\mathcal{M}}_3} = \exp \left[-\frac{i}{4} \alpha \gamma(f \cup f)|_y \right] \langle \mathcal{O}^\alpha \rangle_F, \quad (6.43)$$

where, on the right-hand side, an operator $\mathcal{O}^\alpha$ is given without the defect but by the transformations of the fermion fields

$$\psi(n)^\alpha = e^{-i\frac{\alpha}{2}\delta_{ny}} \psi(n), \quad \bar{\psi}(n)^\alpha = \bar{\psi}(n) e^{i\frac{\alpha}{2}\delta_{ny}}. \quad (6.44)$$

By using the relation (6.43) site by site, we have the anomalous Ward–Takahashi identity for an arbitrary defect $\widetilde{\mathcal{M}}_3 = \partial\mathcal{V}_4$,

$$\langle \mathcal{O} \rangle_F^{\widetilde{\mathcal{M}}_3} = \exp \left[-\frac{i}{4} \alpha \gamma \sum_{\text{hypercube} \in \mathcal{V}_4} f \cup f \right] \langle \mathcal{O}^\alpha \rangle_F, \quad (6.45)$$

where, on the right-hand side,

$$\psi(n)^\alpha = \begin{cases} e^{-i\alpha/2} \psi(n) & \text{for } n \in \mathcal{V}_4, \\ \psi(n) & \text{otherwise.} \end{cases} \quad \bar{\psi}(n)^\alpha = \begin{cases} \bar{\psi}(n) e^{i\alpha/2} & \text{for } n \in \mathcal{V}_4, \\ \bar{\psi}(n) & \text{otherwise.} \end{cases} \quad (6.46)$$

Equation (6.45) implies that the deformation leaves the effect of the axial anomaly as we argued in the continuum as Eq. (6.3); that is, the defect is not topological for this anomalous symmetry. One may rewrite Eq. (6.45) and define the symmetry operator $U_\alpha(\widetilde{\mathcal{M}}_3)$ as

$$\left\langle U_\alpha(\widetilde{\mathcal{M}}_3) \mathcal{O} \right\rangle_F \equiv \langle \mathcal{O} \rangle_F^{\widetilde{\mathcal{M}}_3} \exp \left[\frac{i}{4} \alpha \gamma \sum_{\text{hypercube} \in \mathcal{V}_4} f \cup f \right] = \langle \mathcal{O}^\alpha \rangle_F. \quad (6.47)$$

$U_\alpha(\widetilde{\mathcal{M}}_3)$ is again topological but not invariant under the gauge transformations on *the boundary variables* as we will see below.

To see the boundary structure, now, we introduce the logarithm of the link variable as

$$a_\mu(n) \equiv \frac{1}{i} \ln u(n, \mu), \quad -\pi < a_\mu(n) \leq \pi. \quad (6.48)$$

Then, the field strength can be rewritten by

$$f = \delta a + 2\pi z, \quad (6.49)$$

where $z_{\mu\nu}(n) \in \mathbb{Z}$ (*different from* the 't Hooft flux above) needs to remedy the discrepancy between the choices of the logarithmic branch, and δ is the coboundary operator.¹¹ Equation (6.47) becomes

$$\langle \mathcal{O} \rangle_{\text{F}}^{\widetilde{\mathcal{M}}_3} \exp \left[-\frac{ip}{4\pi N} \sum_{\text{cube} \in \mathcal{M}_3} (a \cup f + 2\pi z \cup a) \right] \exp \left(-\frac{ip\pi}{N} \sum_{\text{hypercube} \in \mathcal{V}_4} z \cup z \right) = \langle \mathcal{O}^\alpha \rangle_{\text{F}}, \quad (6.50)$$

The first exponential factor on the left-hand side would be regarded as a lattice counterpart of the Chern–Simons 3-form. An infinitesimal variation of $u(x, \mu)$ provides a change depending only on the variation of $u(x, \mu)$ on a 3-dimensional closed surface corresponding to $\widetilde{\mathcal{M}}_3$. We denote this closed 3-surface on the original lattice as $\mathcal{M}_3$; see Fig. 4. We call $u_\ell = u_{\langle x \rightarrow x + \hat{\mu} \rangle \in \mathcal{M}_3}$ boundary variables, while $u_{\langle x \rightarrow x + \hat{\mu} \rangle \in \text{int}(\mathcal{M}_3)}$ bulk variables. We thus regard the quantity in the square brackets of Eq. (6.47) as a possible definition of the $U(1)$ Chern–Simons term associated with the defect $\widetilde{\mathcal{M}}_3 = \partial(\mathcal{V}_4)$.

How to make the expression invariant also under the gauge transformation on the boundary variables only is the subject in the following.

6.4.2 Gauge average and the projection operator for magnetic fluxes on the defect

To this end, we first imitate the prescription by Karasik [10] in continuum and so integrate $U_\alpha(\widetilde{\mathcal{M}}_3)$ over gauge transformations on the boundary variables only. That is, we consider the average over gauge transformations:

$$\left\langle \tilde{U}_\alpha(\widetilde{\mathcal{M}}_3) \mathcal{O} \right\rangle_{\text{F}} \equiv \langle \mathcal{O} \rangle_{\text{F}}^{\widetilde{\mathcal{M}}_3} \int \mathcal{D}\lambda \exp \left[\frac{i}{4} \alpha \gamma \sum_{\text{hypercube} \in \mathcal{V}_4} f \cup f \right]^\lambda = \langle \mathcal{O}^\alpha \rangle_{\text{F}}, \quad (6.51)$$

where $\mathcal{D}\lambda$ is the integration over gauge transformations on the boundary variables only. The symbol $[\]^\lambda$ indicates the gauge transformation on the boundary variables only

¹¹ Precisely speaking, a is a 1-cochain in Γ , $a \in C^1(\Gamma; \mathbb{R})$, z is a $\mathbb{Z}$ 2-cocycle in Γ , $z \in Z^2(\Gamma; \mathbb{Z})$, and f is a 2-cocycle, $f \in Z^2(\Gamma; \mathbb{Z})$

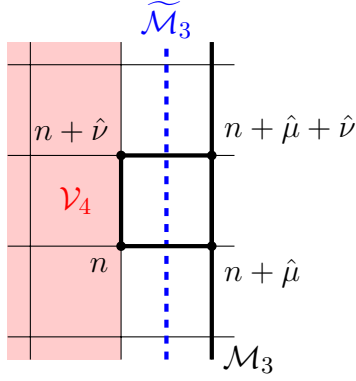


Fig. 6 A plaquette extending between $\mathcal{M}_3$ and $\mathcal{V}_4$. Only the gauge field $u(n + \hat{\mu}, \nu)$ on the link $\langle n + \hat{\mu} \rightarrow n + \hat{\mu} + \hat{\nu} \rangle$ is the boundary variable which is subject of the gauge transformation in Eq. (6.55).

corresponding to the gauge average of the Chern–Simons action on the defect in the continuum [10].

Suppose that, in a plaquette, only u_ℓ on a link $\ell = \langle s \rightarrow s' \rangle$ is the boundary variable, while others are bulk variables as illustrated in Fig. 6. Then, we consider the gauge transformation acting only on the boundary variable,

$$u_\ell \rightarrow \lambda_s u_\ell \lambda_{s'}^{-1} = e^{-i\phi_s} u_\ell e^{i\phi_{s'}}, \quad \text{with } \lambda = e^{-i\phi}, \quad -\pi < \phi \leq \pi. \quad (6.52)$$

To define $\mathcal{D}\lambda$ in the path integral, we have to make the winding number well-defined as we will construct below. Actually, such a definition is given in a ad hoc prescription and *gauge non-invariant* smoothness condition. While the admissibility imposes the constraint on plaquettes, the desired non-invertibility appears to require a “gauge average” for smooth configurations of the *link variables* itself. To realize the picture of Ref. [10], we focus on possible gauge transformations λ as follows: Setting,

$$-\pi < \frac{1}{i} \ln \left(e^{-i\phi_s} e^{i\phi_{s'}} \right) = \partial\phi_\ell + 2\pi l_\ell \leq \pi, \quad (6.53)$$

where $l_\ell \in \mathbb{Z}$, we require sufficiently smooth variations of gauge transformation functions on the lattice as

$$\sup_\ell |\partial\phi_\ell + 2\pi l_\ell| < \varepsilon', \quad 0 < \varepsilon' < \frac{\pi}{6}. \quad (6.54)$$

The transformation (6.52) then induces the change of the field strength,

$$f_{\mu\nu}(n) \mapsto f_{\mu\nu}(n) + \partial_\nu \phi(n + \hat{\mu}) + 2\pi l_\nu(n + \hat{\mu}). \quad (6.55)$$

Note that the right-hand side is within the desired branch because of the admissibility (6.13) and the smoothness condition (6.54). Equation (6.54) also implies that l_ℓ is flat, $\delta l_\ell = 0$.

Even excluding the gauge non-invariance of the smoothness condition, the gauge invariance of the Chern–Simons term is archived, however, in a restricted sense. Under the gauge transformation λ' on boundary variables only,

$$\left\langle \tilde{U}_\alpha(\tilde{\mathcal{M}}_3)\mathcal{O} \right\rangle_{\text{F}}^{\lambda'} = \langle \mathcal{O} \rangle_{\text{F}}^{\tilde{\mathcal{M}}_3} \int \mathcal{D}\lambda \exp \left[\frac{i}{4} \alpha \gamma \sum_{\text{hypercube} \in \mathcal{V}_4} f \cup f \right]^{\lambda \lambda'}. \quad (6.56)$$

At a first sight, one might think that λ' can be absorbed by the redefinition $\lambda \rightarrow \lambda(\lambda')^{-1}$. Is is problematic because the flatness, $\delta l_\ell = 0$, should be kept intact in discussions below, and our measure $\mathcal{D}\lambda$ is not invariant under such a shift. If λ' is sufficiently smooth, $|\partial\phi'| \ll 1$ for $\lambda' = e^{-i\phi'}$, l_ℓ defined does not change under the redefinition. We thus require this sufficient smoothness for λ' in Eq. (6.56). We hope that this point is not physically problematic because in continuum limit only differentiable gauge transformations are relevant.

Let us now fix a direction μ and assume that $\tilde{\mathcal{M}}_3 = T^3$ with the size L is perpendicular to $\hat{\mu}$ and intersects a link $\langle n \rightarrow n + \hat{\mu} \rangle$ as Fig. 6.

Note that the flatness, $\delta l = 0$, allows us to introduce a “scalar potential” $\varphi(n)$ as

$$l_{\ell=\langle n \rightarrow n+\hat{\nu} \rangle} = \partial_\nu \varphi(n), \quad \varphi(n) = \sum_{\langle y \rightarrow y+\hat{\rho} \rangle \in \text{a path connecting } o \text{ and } n \text{ in } \mathcal{M}_3} l_\rho(y), \quad (6.57)$$

where we have assumed a certain fixed point $o \in \mathcal{M}_3$. $\varphi(x)$ is topological under any deformation of paths but is not single-valued in $\mathcal{M}_3$ when the path winds on three non-trivial cycles of $\mathcal{M}_3$. Under a single winding to the ν direction, $\varphi(x)$ acquires an additional integer

$$k_\nu = \sum_{s=0}^{L-1} l_\nu(n + s\hat{\nu}) = \sum_{s=0}^{L-1} \left[\frac{1}{2\pi} \partial_\nu \phi(n + s\hat{\nu}) + l_\nu(n + s\hat{\nu}) \right]. \quad (6.58)$$

This number is *independent* of the site n in $\mathcal{M}_3$. One sees that k_ν is the number of how many times $e^{-i\phi(x)}e^{+i\phi(x+\hat{\nu})}$ winds around $U(1)$ as n goes along the ν direction. The above smoothness condition is required to make this winding number well-defined on the lattice. Therefore, if we integrate over possible smooth gauge transformations in Eq. (6.51), one has a summation over integers k_ν for each ν . Because of Eq. (6.54), for finite L , the winding number is bounded by

$$|k_\nu| < \frac{\epsilon'}{2\pi} L. \quad (6.59)$$

Finally, using the representation (6.57) in $f \cup f$ we have the combination around the ν direction as

$$\pi k_\nu \sum_{n+\hat{\mu} \in \mathcal{M}_3, x_\nu=0} \epsilon_{\mu\nu\rho\sigma} [f_{\rho\sigma}(n + \hat{\mu}) + f_{\rho\sigma}(n)], \quad (6.60)$$

Therefore, we obtain

$$\int \mathcal{D}\lambda \exp \left[8\pi i \alpha \gamma k_\nu \sum_{n \in \mathcal{M}_2^\nu} \epsilon_{\mu\nu\rho\sigma} f_{\rho\sigma}(n) \right], \quad (6.61)$$

where $\mathcal{M}_2^\nu$ is a closed surface such that on $\mathcal{M}_3 = S^1 \times \mathcal{M}_2^\nu$, S^1 is the direction of ν . Integration over smooth gauge transformations would become the sum over the winding number k_ν . Introducing the function $\delta_L(x)$ by

$$\sum_{k=-[\varepsilon' L/(2\pi)]}^{[\varepsilon' L/(2\pi)]} e^{ikx} = 2\pi \sum_{n=-\infty}^{\infty} \delta_L(x - 2\pi n), \quad (6.62)$$

we know $\delta_L(x) \xrightarrow{\ell \rightarrow \infty} \delta(x)$. Therefore, the gauge average gives rise to, for each ν ,

$$\delta_L \left(\frac{\alpha}{2\pi} \frac{1}{4\pi} \sum_{n \in \mathcal{M}_2^\nu} \epsilon_{\mu\nu\rho\sigma} f_{\rho\sigma}(n) - \mathbb{Z} \right), \quad (6.63)$$

This implies that

$$\frac{1}{4\pi} \sum_{n \in \mathcal{M}_2^\nu} \epsilon_{\mu\nu\rho\sigma} f_{\rho\sigma}(n) \xrightarrow{L \rightarrow \infty} \begin{cases} 0 & \alpha/(2\pi) \text{ is an irrational number,} \\ N\mathbb{Z} & \alpha/(2\pi) \text{ is a rational number, } p/N, \end{cases} \quad (6.64)$$

where p and N are co-prime integers. This precisely corresponds to the constraint on the magnetic flux in the continuum obtained in Ref. [10]: $\frac{1}{2\pi} \int_{\mathcal{M}_2} da \in N\mathbb{Z}$. In the infinite volume, $L \rightarrow \infty$, the symmetry operator can be defined in terms of Eq. (6.51) as

$$\tilde{U}_\alpha(\widetilde{\mathcal{M}}_3) = U_\alpha(\widetilde{\mathcal{M}}_3) P_\alpha(\widetilde{\mathcal{M}}_3), \quad (6.65)$$

where $P_\alpha(\widetilde{\mathcal{M}}_3)$ is a projection operator on the subspace of allowed magnetic fluxes in $\widetilde{\mathcal{M}}_3$. From this expression, it is obvious that $\tilde{U}_\alpha(\widetilde{\mathcal{M}}_3)$ possesses the non-invertibility.

6.4.3 Symmetry operator and fusion rules in terms of a $\mathbb{Z}_N$ TQFT

We obtained the non-invertible symmetry operator by Karasik's prescription, while there are some subtleties in our lattice formulation. One may wonder whether those have some

physical and relevant meanings or not, and are lattice artifacts or a kind of no-go. The construction also refers to an auxiliary 4-dimensional volume on the lattice, but most people want to achieve an intrinsically 3-dimensional construction.

We initially set the rotation angle as $\alpha = 2\pi p/N$ by integers p and N , which are co-prime each other. We aim to employ a 3-dimensional $\mathbb{Z}_N$ TQFT, the level- N BF theory [21, 23, 33, 54, 89, 90]. On the basis of our lattice formulation coupled to the level- N BF theory, we can impose desired properties: gauge invariance, topologicalness, and non-invertibility; and also we can give the evaluation of fusion rules within our lattice regularized framework. Our formulation from here is possible only when p is an even integer $p = 2\mathbb{Z}$.

In the anomalous Ward–Takahashi identity (6.50), the first exponential on the left-hand side has the form of a 3-dimensional sum and this part can be generalized to arbitrary, i.e., not necessarily homologically trivial, closed 3-dimensional surfaces. The problem is the second exponential because it refers to a 4-dimensional volume $\mathcal{V}_4$. To express this solely in terms of 3-dimensional notions, we introduce the level- N BF theory defined on a 3-dimensional closed surface (i.e., 3-cycle) $\mathcal{M}_3$ in Γ by

$$\begin{aligned} S_{\text{BF}} &= \frac{-ip\pi}{N} \sum_{\text{cube} \in \mathcal{M}_3} [b(\delta c - z) - z \cup c] \\ &= \frac{-ip\pi}{N} \sum_{n \in \mathcal{M}_3} \epsilon_{\mu\nu\rho} \left\{ b_\mu(\tilde{n}) \left[\partial_\nu c_\rho(n + \hat{\mu}) - \frac{1}{2} z_{\nu\rho}(n + \hat{\mu}) \right] - \frac{1}{2} z_{\mu\nu}(n) c_\rho(n + \hat{\mu} + \hat{\nu}) \right\}, \end{aligned} \quad (6.66)$$

where we put the $\mathbb{Z}_N$ 1-cochain c on Γ and the $\mathbb{Z}_N$ 1-cochain b on the dual lattice,¹² whose sites are given by

$$\tilde{n} = n + \frac{1}{2}\hat{\mu} + \frac{1}{2}\hat{\nu} + \frac{1}{2}\hat{\rho}, \quad (6.67)$$

assuming that $\mathcal{M}_3$ extends in μ, ν, ρ directions. The first term of the action (6.66) has the structure depicted in Fig. 7 (*not* cup product).

The partition function of the BF theory is then defined by

$$\mathcal{Z}_{\mathcal{M}_3}[z] = \frac{1}{N^s} \int \mathcal{D}b \mathcal{D}c e^{-S_{\text{BF}}}, \text{ with } \mathcal{D}b \equiv \prod_{\mu, n \in \mathcal{M}_3} \frac{1}{N} \sum_{b_\mu(\tilde{n})=0}^{N-1}, \mathcal{D}c \equiv \prod_{\mu, n \in \mathcal{M}_3} \sum_{c_\mu(n)=0}^{N-1}. \quad (6.68)$$

where the partition function is divided by the gauge volume $|C^0(\mathcal{M}_3; \mathbb{Z}_N)| = N^s$, where s denotes the number of sites in $\mathcal{M}_3$. Note that for p even $e^{-S_{\text{BF}}}$ is invariant under $b \mapsto b + N\mathbb{Z}$ and $c \mapsto c + N\mathbb{Z}$, which is consistent with the summation in Eq. (6.68). Also, since

¹² $b \in C^1(\widetilde{\mathcal{M}}_3; \mathbb{Z}_N)$ and $c \in C^1(\mathcal{M}_3; \mathbb{Z}_N)$

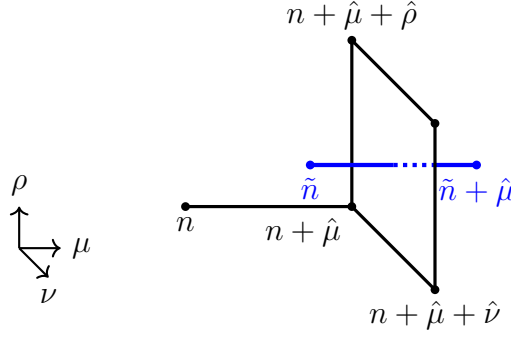


Fig. 7 The structure of the product in the first term of the lattice action (6.66). The 1-cochain b is put on the dual link (the blue line) and the 1-cochain c is put on the original lattice (the black lines); 2-cochain $\delta c - z$ is therefore put on the original plaquette (the square).

$\delta z = 0$, $e^{-S_{\text{BF}}}$ is invariant under the 0-form $\mathbb{Z}_N$ gauge transformation, $c \rightarrow c + \delta\mu$, where $\mu \in C^0(\mathcal{M}_3; \mathbb{Z}_N)$.

Now, taking the sum over b in Eq. (6.68), for p even, we have

$$\mathcal{Z}_{\mathcal{M}_3}[z] = \frac{1}{N^s} \int \mathcal{D}c \delta_N [\delta c - z] \exp \left(-\frac{ip\pi}{N} \sum_{\text{cube} \in \mathcal{M}_3} z \cup c \right), \quad (6.69)$$

where we have introduced the delta functional:

$$\delta_N [\delta c - z] \equiv \prod_{\mu, n \in \mathcal{M}_3, \nu \neq \mu, \rho \neq \mu, \nu < \rho} \delta_{\partial_\nu c_\rho(n+\hat{\mu}) - \partial_\rho c_\nu(n+\hat{\mu}) - z_{\nu\rho}(n+\hat{\mu}), N\mathbb{Z}}. \quad (6.70)$$

Equation (6.69) immediately shows that, if there exists a 2-dimensional closed surface $\mathcal{M}_2 \subset \mathcal{M}_3$ such that the magnetic flux is non-zero modulo N , i.e.,

$$\sum_{\text{face} \in \mathcal{M}_2} z = \frac{1}{2} \sum_{\mu, \nu} \sum_{n \in \mathcal{M}_2} \epsilon_{\mu\nu} z_{\mu\nu}(n) \neq 0 \bmod N, \quad (6.71)$$

then there is no configuration of c such that $\delta c = z \bmod N$. In this case, therefore, the partition function identically vanishes, $\mathcal{Z}_{\mathcal{M}_3}[z] = 0$. Later, we substitute $z \rightarrow z|_{\mathcal{M}_3}$, where $z|_{\mathcal{M}_3}$ denotes the 2-cocycle $z \in Z^2(\Gamma, \mathbb{Z})$ in Eq. (6.50) restricted on $\mathcal{M}_3$. If this $z|_{\mathcal{M}_3}$ satisfies Eq. (6.71) (this is equivalent to $[1/(2\pi)] \sum_{\text{face} \in \mathcal{M}_2} f \neq 0 \bmod N$), we cannot construct the symmetry operator. The existence of the symmetry operator, therefore, requires that the magnetic fluxes in Eq. (6.14) are quantized in unit of N .

Let us first set $z = 0$ in Eq. (6.69). Then, the partition function is evaluated as

$$\mathcal{Z}_{\mathcal{M}_3}[0] = \frac{1}{N^s} \int \mathcal{D}c \delta_N [\delta c] = N^{b_2-1}, \quad (6.72)$$

where $b_2 = |H^2(\mathcal{M}_3; \mathbb{Z}_N)|$ is the second Betti number of $\mathcal{M}_3$. Note that, letting numbers of the sites, links, faces (plaquettes), and cubes of $\mathcal{M}_3$, be s , l , f , and c , respectively, the Euler number vanishes for 3-cycles, $s - l + f - c = \chi(\mathcal{M}_3) = 0$. Therefore, with the definition in Eq. (6.68), the partition function $\mathcal{Z}_{\mathcal{M}_3}[0]$ is topological, depending only on the topology of $\mathcal{M}_3$. For instance, $b_2 = 0$ for S^3 , $b_2 = 1$ for $S^2 \times S^2$, and $b_2 = 3$ for T^3 .

Next, let us consider the case of $z \neq 0$, that is, $z \bmod N = 0$ in $H^2(\mathcal{M}_3; \mathbb{Z}_N)$. Then, there exists $\nu \in C^1(\mathcal{M}_3; \mathbb{Z}_N)$ such that

$$z = \delta\nu \bmod N. \quad (6.73)$$

Substituting this into Eq. (6.69), we have

$$\mathcal{Z}_{\mathcal{M}_3}[z] = \exp \left(-\frac{ip\pi}{N} \sum_{\text{cube} \in \mathcal{M}_3} \delta\nu \cup \nu \right) \mathcal{Z}_{\mathcal{M}_3}[0], \quad (6.74)$$

where we have shifted c as $c \rightarrow c + \nu \bmod N$ and then used $\delta\nu \cup c = \delta(\nu \cup c) + \nu \cup \delta c$. Finally, we have used $\delta c = 0 \bmod N$ which holds under the delta functional after the shift.

We now consider a deformation of the 3-cycle $\mathcal{M}_3 \rightarrow \mathcal{M}'_3$ within Γ . After some calculations, one obtains

$$\mathcal{Z}_{\mathcal{M}'_3}[z|\mathcal{M}'_3] = \exp \left(-\frac{ip\pi}{N} \sum_{\text{hypercube} \in \mathcal{V}_4} z \cup z \right) \mathcal{Z}_{\mathcal{M}_3}[z|\mathcal{M}_3], \quad (6.75)$$

where $\partial(\mathcal{V}_4) = \mathcal{M}'_3 \cap (-\mathcal{M}_3)$. Comparing this with Eq. (6.50), which depends on the 4-dimensional explanation as $z \cup z$, we arrive at the conclusion that the symmetry operator on an *arbitrary* 3-cycle $\widetilde{M}_3$ is given by using the BF theory as

$$\left\langle \tilde{U}_{2\pi p/N}(\widetilde{M}_3) \mathcal{O} \right\rangle_{\text{F}} = \langle \mathcal{O} \rangle_{\text{F}}^{\widetilde{M}_3} \exp \left[-\frac{ip}{4\pi N} \sum_{\text{cube} \in \mathcal{M}_3} (a \cup f + 2\pi z \cup a) \right] \mathcal{Z}_{\mathcal{M}_3}[z|\mathcal{M}_3]. \quad (6.76)$$

Under the deformation of the defect $\widetilde{M}_3 \rightarrow \widetilde{M}'_3$, obviously we have

$$\left\langle \tilde{U}_{2\pi p/N}(\widetilde{M}'_3) \mathcal{O} \right\rangle_{\text{F}} = \left\langle \tilde{U}_{2\pi p/N}(\widetilde{M}_3) \mathcal{O} \right\rangle_{\text{F}}, \quad (6.77)$$

where we have used the anomalous Ward–Takahashi identity (6.50). Therefore this shows that the symmetry operator is actually topological. One can find that the symmetry operator

is also manifestly invariant under arbitrary 3-dimensional lattice gauge transformations; from the perspective of anomaly inflow, the action described by $z \cup z$ is a 4-dimensional SPT action.

As an application of our representation (6.76), we evaluate fusion rules of two symmetry operators sharing a common 3-cycle $\mathcal{M}_3$. From Eq. (6.69), we have (here we assume that $\gcd(p_1 + p_2, N) = 1$)

$$\begin{aligned} \mathcal{Z}_{\mathcal{M}_3}^{(p_1, N)}[z] \mathcal{Z}_{\mathcal{M}_3}^{(p_2, N)}[z] &= \frac{1}{N^{2s}} \int \mathcal{D}c_1 \mathcal{D}c_2 \delta_N [\delta c_1 - z] \delta_N [\delta c_2 - z] \\ &\quad \times \exp \left[-\frac{ip_1 \pi}{N} \sum_{\text{cube} \in \mathcal{M}_3} z \cup (c_1 - c_2) - \frac{i(p_1 + p_2) \pi}{N} \sum_{\text{cube} \in \mathcal{M}_3} z \cup c_2 \right] \\ &= \mathcal{Z}_{\mathcal{M}_3}[0] \mathcal{Z}_{\mathcal{M}_3}^{(p_1 + p_2, N)}[z]. \end{aligned} \quad (6.78)$$

Hence, the fusion rule of the symmetry operators reads

$$\tilde{U}_{2\pi p_1/N} \tilde{U}_{2\pi p_2/N} = \mathcal{Z}_{\mathcal{M}_3}[0] \tilde{U}_{2\pi(p_1 + p_2)/N}. \quad (6.79)$$

The fact that this fusion depends on the partition function of a TQFT, $\mathcal{Z}_{\mathcal{M}_3}[0]$ (6.72), shows the non-invertibility.

Setting $p_2 = -p_1 = p$ in Eq. (6.78), we have the condensation operator $\mathcal{C}_{\mathcal{M}_3}[z]$,

$$\mathcal{Z}_{\mathcal{M}_3}^{(p, N)}[z] \mathcal{Z}_{-\mathcal{M}_3}^{(p, N)}[z] = \mathcal{Z}_{\mathcal{M}_3}^{(p, N)}[z] \mathcal{Z}_{\mathcal{M}_3}^{(-p, N)}[z] = \mathcal{Z}_{\mathcal{M}_3}[0] \mathcal{C}_{\mathcal{M}_3}[z], \quad (6.80)$$

where

$$\mathcal{C}_{\mathcal{M}_3}[z] = \mathcal{Z}_{\mathcal{M}_3}^{(0, N)}[z] = \begin{cases} \mathcal{Z}_{\mathcal{M}_3}[0] & \text{for } z = 0 \bmod N \in H^2(\mathcal{M}_3; \mathbb{Z}_N), \\ 0 & \text{otherwise.} \end{cases} \quad (6.81)$$

The condensation operator $\mathcal{C}_{\mathcal{M}_3}[z]$ is equal to the topological factor $\mathcal{Z}_{\mathcal{M}_3}[0]$ with the projection operator which restricts the z flux to multiples of N .

A more general fusion can be obtained by considering (here we assume that $\gcd(p_1, N_1) = \gcd(p_2, N_2) = 1$)

$$\begin{aligned} \mathcal{Z}_{\mathcal{M}_3}^{(p_1, N_1)}[z] \mathcal{Z}_{\mathcal{M}_3}^{(p_2, N_2)}[z] &= \frac{1}{N^{2s}} \int \mathcal{D}c_1 \mathcal{D}c_2 \delta_{N_1} [\delta c_1] \delta_{N_2} [\delta c_2] \exp \left[-\left(\frac{ip_1 \pi}{N_1} + \frac{ip_2 \pi}{N_2} \right) \sum_{\text{cube} \in \mathcal{M}_3} \delta \nu \cup \nu \right], \end{aligned} \quad (6.82)$$

where we have shifted the variables, $c_1 \rightarrow c_1 + \nu$ and $c_2 \rightarrow c_2 + \nu$, such that $z = \delta \nu \bmod N = \text{lcm}(N_1, N_2)$. On the other hand, setting $\beta_1 = N/N_1$ and $\beta_2 = N/N_2$ for $N = \text{lcm}(N_1, N_2)$,

and assuming that $\gcd(\beta_1 p_1 + \beta_2 p_2, N) = 1$,

$$\mathcal{Z}_{\mathcal{M}_3}^{(\beta_1 p_1 + \beta_2 p_2, N)}[z] = \frac{1}{N^s} \int \mathcal{D}c \delta_N [\delta c - z] \exp \left[- \left(\frac{i p_1 \pi}{N_1} + \frac{i p_2 \pi}{N_2} \right) \sum_{\text{cube} \in \mathcal{M}_3} z \cup c \right]. \quad (6.83)$$

Therefore, we have the fusion,

$$\mathcal{Z}_{\mathcal{M}_3}^{(p_1, N_1)}[z] \mathcal{Z}_{\mathcal{M}_3}^{(p_2, N_2)}[z] = \frac{\mathcal{Z}_{\mathcal{M}_3}^{(0, N_1)}[0] \mathcal{Z}_{\mathcal{M}_3}^{(0, N_2)}[0]}{\mathcal{Z}_{\mathcal{M}_3}^{(0, N)}[0]} \mathcal{Z}_{\mathcal{M}_3}^{(\beta_1 p_1 + \beta_2 p_2, N)}[z], \quad (6.84)$$

and this shows that

$$\tilde{U}_{2\pi p_1/N_1} \tilde{U}_{2\pi p_2/N_2} = \frac{\mathcal{Z}_{\mathcal{M}_3}^{(0, N_1)}[0] \mathcal{Z}_{\mathcal{M}_3}^{(0, N_2)}[0]}{\mathcal{Z}_{\mathcal{M}_3}^{(0, N)}[0]} \tilde{U}_{2\pi(p_1/N_1 + p_2/N_2)}. \quad (6.85)$$

From Eq. (6.72), the overall factor in the right-hand side can be written by $(N_1 N_2 / N)^{b_2 - 1}$.

A Counterterm in continuum theory

In this Appendix, we illustrate an issue in chiral gauge theory with an anomalous matter content, that is, the definition of the gauge current [91]. This analysis is important to include what sort of counterterm in our lattice formulation in the main text. We consider the same system, the $U(1) \times U(1)'$ gauge theory with left-handed Weyl fermions. Hence, we introduce the Dirac operator by

$$\mathcal{D} = \gamma_\mu D_\mu = \gamma_\mu (\partial_\mu + i a_\mu t + i A_\mu T), \quad (A1)$$

where we used Eq. (6.32).

The effective action $\Gamma \equiv -\ln \int \mathcal{D}\psi \mathcal{D}\bar{\psi} e^{-\int d^4x \bar{\psi} \mathcal{D} \psi}$, a functional of a_μ and A_μ , is defined in the following [92]: First, we note

$$\begin{aligned} \Gamma &= \int_0^1 ds \frac{d}{ds} \Gamma|_{a_\mu \rightarrow s a_\mu, A_\mu \rightarrow s A_\mu} \\ &= \int_0^1 ds \frac{1}{s} \int d^4x \left[a_\mu(x) \frac{\delta}{\delta a_\mu(x)} \Gamma + A_\mu(x) \frac{\delta}{\delta A_\mu(x)} \Gamma \right] \Big|_{a_\mu \rightarrow s a_\mu, A_\mu \rightarrow s A_\mu} \\ &\equiv i \int_0^1 ds \frac{1}{s} \int d^4x \left[a_\mu(x) \tilde{\ell}_\mu(x) + A_\mu(x) \tilde{L}_\mu(x) \right] \Big|_{a_\mu \rightarrow s a_\mu, A_\mu \rightarrow s A_\mu}, \end{aligned} \quad (A2)$$

where $\tilde{\ell}_\mu(x)$ and $\tilde{L}_\mu(x)$ are gauge currents. This *formal* introduction suffers from UV divergences; we thus define the gauge currents $\tilde{\ell}_\mu(x)$ and $\tilde{L}_\mu(x)$ with the UV cutoff Λ as the

covariant gauge current [93],

$$\begin{aligned}\tilde{l}_\mu(x) &= -\text{tr} \left(P_+ \gamma_\mu t \frac{1}{\not{D}} e^{\not{D}^2/\Lambda^2} \right) \delta(x-y)|_{y \rightarrow x}, \\ \tilde{L}_\mu(x) &= -\text{tr} \left(P_+ \gamma_\mu T \frac{1}{\not{D}} e^{\not{D}^2/\Lambda^2} \right) \delta(x-y)|_{y \rightarrow x}.\end{aligned}\tag{A3}$$

The definition of the effective action through Eqs. (A2) and (A3) is known as the covariant regularization [92]. It is shown [94] that the lattice formulation of the $U(1)$ chiral gauge theory [13], at least in infinite volume, can be understood as the covariant regularization. Therefore, it is natural to consider anomalies by using the covariant regularization in the main text.

On the other hand, the consistent gauge current [95, 96] can be defined by the functional derivative of the effective action Γ with respect to the gauge potential as

$$\begin{aligned}\ell_\mu(x) &= -i \frac{\delta}{\delta a_\mu(x)} \Gamma \\ &= \tilde{\ell}_\mu(x) + \int_0^1 ds \int d^4y \ a_\nu(y) \left[\frac{\delta}{\delta a_\mu(x)} \tilde{\ell}_\nu(y) - \frac{\delta}{\delta a_\nu(y)} \tilde{\ell}_\mu(x) \right] \Big|_{a_\mu \rightarrow s a_\mu, A_\mu \rightarrow s A_\mu} \\ &\quad + \int_0^1 ds \int d^4y \ A_\nu(y) \left[\frac{\delta}{\delta a_\mu(x)} \tilde{L}_\nu(y) - \frac{\delta}{\delta A_\nu(y)} \tilde{\ell}_\mu(x) \right] \Big|_{a_\mu \rightarrow s a_\mu, A_\mu \rightarrow s A_\mu},\end{aligned}\tag{A4}$$

and

$$\begin{aligned}L_\mu(x) &= -i \frac{\delta}{\delta A_\mu(x)} \Gamma \\ &= \tilde{L}_\mu(x) + \int_0^1 ds \int d^4y \ A_\nu(y) \left[\frac{\delta}{\delta A_\mu(x)} \tilde{L}_\nu(y) - \frac{\delta}{\delta A_\nu(y)} \tilde{L}_\mu(x) \right] \Big|_{a_\mu \rightarrow s a_\mu, A_\mu \rightarrow s A_\mu} \\ &\quad + \int_0^1 ds \int d^4y \ a_\nu(y) \left[\frac{\delta}{\delta A_\mu(x)} \tilde{\ell}_\nu(y) - \frac{\delta}{\delta a_\nu(y)} \tilde{L}_\mu(x) \right] \Big|_{a_\mu \rightarrow s a_\mu, A_\mu \rightarrow s A_\mu}.\end{aligned}\tag{A5}$$

Thus, the last two terms in Eqs. (A4) and (A5) provide the Bardeen–Zumino current [91], which is the difference between the consistent current and the covariant current. To obtain explicit forms of the Bardeen–Zumino currents, by using the prescription of the covariant regularization, one finds ($\Lambda \rightarrow \infty$ is appropriately understood in what follows) [92, 97].

$$\begin{aligned}\int d^4y \ a_\nu(y) \left[\frac{\delta}{\delta a_\mu(x)} \tilde{\ell}_\nu(y) - \frac{\delta}{\delta a_\nu(y)} \tilde{\ell}_\mu(x) \right] &= -8\gamma \epsilon_{\mu\nu\rho\sigma} a_\nu(x) F_{\rho\sigma}(x), \\ \int d^4y \ A_\nu(y) \left[\frac{\delta}{\delta a_\mu(x)} \tilde{L}_\nu(y) - \frac{\delta}{\delta A_\nu(y)} \tilde{\ell}_\mu(x) \right] &= -8\gamma \epsilon_{\mu\nu\rho\sigma} A_\nu(x) f_{\rho\sigma}(x), \\ \int d^4y \ A_\nu(y) \left[\frac{\delta}{\delta A_\mu(x)} \tilde{L}_\nu(y) - \frac{\delta}{\delta A_\nu(y)} \tilde{L}_\mu(x) \right] &= -8\gamma \epsilon_{\mu\nu\rho\sigma} A_\nu(x) F_{\rho\sigma}(x),\end{aligned}\tag{A6}$$

and

$$\int d^4y a_\nu(y) \left[\frac{\delta}{\delta A_\mu(x)} \tilde{\ell}_\nu(y) - \frac{\delta}{\delta a_\nu(y)} \tilde{L}_\mu(x) \right] = -8\gamma\epsilon_{\mu\nu\rho\sigma} a_\nu(x) f_{\rho\sigma}(x). \quad (\text{A7})$$

Substituting these into Eqs. (A4) and (A5), we have the relation between the consistent and covariant gauge currents,

$$\begin{aligned} \ell_\mu(x) &= \tilde{\ell}_\mu(x) - \frac{8}{3}\gamma\epsilon_{\mu\nu\rho\sigma} [a_\nu(x)F_{\rho\sigma}(x) + A_\nu(x)f_{\rho\sigma}(x)], \\ L_\mu(x) &= \tilde{L}_\mu(x) - \frac{8}{3}\gamma\epsilon_{\mu\nu\rho\sigma} [a_\nu(x)f_{\rho\sigma}(x) + A_\nu(x)F_{\rho\sigma}(x)]. \end{aligned} \quad (\text{A8})$$

Now, the covariant gauge currents (A3) give rise to the covariant gauge anomaly [93]:

$$\begin{aligned} \partial_\mu \tilde{\ell}_\mu(x) &= 2\gamma\epsilon_{\mu\nu\rho\sigma} [f_{\mu\nu}(x)F_{\rho\sigma}(x) + F_{\mu\nu}(x)f_{\rho\sigma}(x)], \\ \partial_\mu \tilde{L}_\mu(x) &= 2\gamma\epsilon_{\mu\nu\rho\sigma} [f_{\mu\nu}(x)f_{\rho\sigma}(x) + F_{\mu\nu}(x)F_{\rho\sigma}(x)]. \end{aligned} \quad (\text{A9})$$

Hence, the consistent gauge currents (A8) provide the consistent gauge anomaly [95, 96]:

$$\begin{aligned} \partial_\mu \ell_\mu(x) &= \frac{2}{3}\gamma\epsilon_{\mu\nu\rho\sigma} [f_{\mu\nu}(x)F_{\rho\sigma}(x) + F_{\mu\nu}(x)f_{\rho\sigma}(x)], \\ \partial_\mu L_\mu(x) &= \frac{2}{3}\gamma\epsilon_{\mu\nu\rho\sigma} [f_{\mu\nu}(x)f_{\rho\sigma}(x) + F_{\mu\nu}(x)F_{\rho\sigma}(x)]. \end{aligned} \quad (\text{A10})$$

In Eq. (A10), the “vector” current $\ell_\mu(x)$ does not conserve, and the overall factor in $\partial_\mu L_\mu(x)$ is smaller by 1/3 than the expected axial anomaly in the target theory. To remedy these two issues, it is known that there is an appropriate counterterm to the effective action. The required counterterm in this case is

$$-i\Delta\Gamma = \frac{8}{3}\gamma \int d^4x \epsilon_{\mu\nu\rho\sigma} A_\mu(x) a_\nu(x) f_{\rho\sigma}(x). \quad (\text{A11})$$

Then we obtain the desired form of the consistent gauge anomalies:

$$\begin{aligned} \partial_\mu \left[\ell_\mu(x) - i\frac{\delta}{\delta a_\mu(x)} \Delta\Gamma \right] &= 0, \\ \partial_\mu \left[L_\mu(x) - i\frac{\delta}{\delta A_\mu(x)} \Delta\Gamma \right] &= 2\gamma\epsilon_{\mu\nu\rho\sigma} f_{\mu\nu}(x) f_{\rho\sigma}(x) + \frac{2}{3}\gamma\epsilon_{\mu\nu\rho\sigma} F_{\mu\nu}(x) F_{\rho\sigma}(x). \end{aligned} \quad (\text{A12})$$

Equation (A11) indicates that a lattice counterpart of counterterm should be added in the effective action in our lattice formulation.

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